World Journal of Microbiology and Biotechnology (2025) 41:278
https://doi.org/10.1007/s11274-025-04494-5

REVIEW

AI-enhanced bioprocess technologies: machine learning implementations from upstream to downstream operations

Deepankar Sharma[1] · **Kavita Singh**[2]

Received: 11 February 2025 / Accepted: 10 July 2025 / Published online: 28 July 2025
© The Author(s), under exclusive licence to Springer Nature B.V. 2025

Abstract

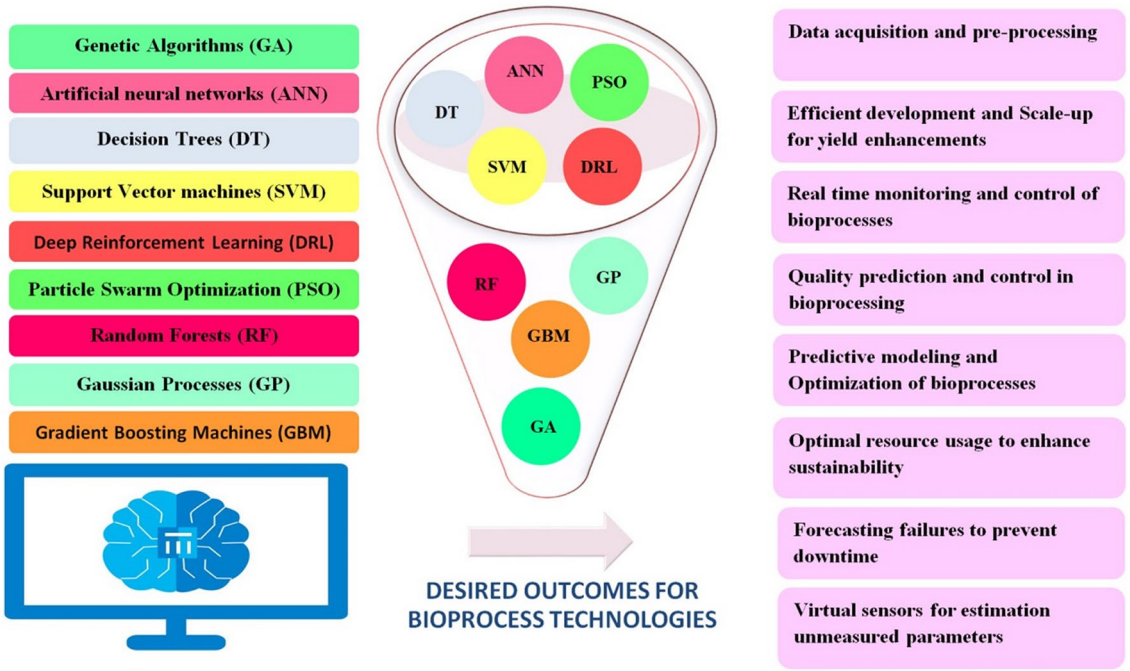
Industrial bioprocesses have forged ahead in recent decades by harnessing the unmatched potential of microorganisms for bioenergy, biochemicals, pharmaceuticals, and food sectors. Any bioprocess technology involves complex upstream and downstream operations, which effectively generate large volumes of complex information across each stage of unit operation. The incorporation of machine learning practices in bioprocess technologies is one of the paradigm shifts that has led to recent advancements. The diverse ML algorithms possess the outstanding capabilities to harness the non-linear relationships and meaningful patterns that are incompletely interpreted using conventional methods. The present review is an up-to-date, in-depth approach that starts with the utilization of machine learning in upstream and downstream operations of bioprocesses. This is followed by describing the methodology of a typical ML workflow, various ML algorithms available and the real-world examples of ML applications that are reported for bioprocess technologies. The major areas covered include the utilization of ML for enhancing the biofuel and biorefinery bioprocesses, pharmaceutical bioprocesses, biochemicals production and optimization, fermented food and beverages bioprocess technologies and the use of ML for monitoring and control of bioprocesses. Moreover, the patents registered on the use of machine learning in bioprocess development are also discussed. The present review provides valuable suggestions for the deployment of diverse ML methodologies for enhancing the purity, yield and productivity of bioprocesses, soft sensors development and utilization of hybrid modeling approach for enhancing the ML- guided monitoring and control of bioprocesses.

Keywords Machine learning · Bioprocess technologies · Patents · Bioprocess optimization · Real-time monitoring · Quality prediction and control

✉ Kavita Singh
kavya.hcst@gmail.com

[1] Department of Biotechnology, O.P. Jindal University (OPJU), Raigarh, Chhattisgarh 496109, India

[2] Department of Biosciences & Biomedical Engineering, Indian Institute of Technology Indore, Indore 453552, India



Introduction

The exploding population growth throughout the world is causing a tremendous surge in demand to produce more food, chemicals, pharmaceuticals, fuels, and other essential goods. The population has reached 7.7 billion and is expected to reach 9.7 billion in 2050 (Zheng et al. 2021). Moreover, in response to recent events of the pandemic and climate change, governing bodies around the world are emphasising advancements in self-reliance, circular economy and sustainable industrial growth (Mikulčić et al. 2023). Bioprocess industries have made a significant contribution to fulfilling these demands by providing diverse, sustainable microbial products (Boodhoo et al. 2022; Usmani et al. 2021). Bioprocess technologies involve the usage of living cells or their parts to create biofuels, biochemicals, food, beverages, biomaterials, and healthcare products (Patel and Shah 2022). Thus, bioprocessing is crucial for addressing the societal needs and preserving the environment. However, there is a continuous need for advancements in biomanufacturing practices to meet the goals of the more demanding future.

The complete bioprocess for the production of a desired product in a bioprocessing industry includes the upstream and downstream operations. Upstream processing involves the inoculum preparation and full-scale cultivation of cells, while downstream processing is associated with the isolation and purification of the desired final product. These complex upstream and downstream operations effectively generate large volumes of data, both in real-time (online) and offline conditions across each stage of unit operation (Rathore et al. 2021; Ündey et al. 2010). This data is obtained from various sources, namely the process parameters regulating the whole bioprocess, bioreactor operation, control systems, analytical sensor data, omics data (proteomics, genomics and metabolomics), material inputs, and process outputs. In any bioprocess technology operation, a large number of parameters exist that influence or control the whole process, and a complex non-linear relationship exists between the process variables and the outcomes. These non-linear relationships and meaningful patterns are very difficult to interpret or are incompletely interpreted using conventional methods like the one-factor-at-a-time (OFAT) approach or the basic design of experiments. OFAT approach assumes that variables act independently, while the basic design of experiments (factorial designs) assumes low-order polynomial relationships. Therefore, many bioprocess technologies often face diverse challenges that can lead to low conversion rates. The final outcome is the unrealistic determination of optimal conditions resulting in low yield and productivity (de Oliveira et al. 2018; Gottumukkala et al. 2017). Several other problems exist in conventional bioprocessing, viz., inconsistent production titres and complex assessment of the environmental impact of bioprocessing activities (Gerstweiler et al. 2021). Also, another significant challenge in bioprocesses is the scale-up effect, in which increasing the scale from the laboratory to the industrial level often led to poorer performance and thus, reduces the economic benefits (de Mello et al. 2024; Wang et al. 2020). Therefore, optimizing and controlling the bioprocesses at different scales

is essential to improve performance and to counter negative effects arising from complex conditions.

Artificial intelligence (AI) is a discipline of computer science that emphasizes the creation of intelligent machines that can efficiently execute the tasks that require human intelligence. AI programs have the capability to autonomously make decisions by following preset rules or recognizable patterns in data (Sarker 2022). Machine learning (ML) is a section of AI that teaches computers to learn from data and continuously upgrade their performance over time. The developers carry out the upgrade through continuous monitoring of the ML models and update them with the available data. Thus, ML algorithms typically analyze historical data to identify patterns and relationships and empower them to make predictive decisions on new data (Holzinger et al. 2023; Goodswen et al. 2021). Different machine learning algorithms tackle various tasks and process data to make decisions. These tasks include classification, which involves sorting of the data into predefined groups, regression analysis for predicting the numerical outcomes, clustering for grouping similar data together, dimensionality reduction for simplifying data and retaining essential information, anomaly detection for spotting the outliers and Reinforcement learning to make sequential decisions (Sarker 2021). In recent times, there has been an increase in the utilization of ML technologies for optimizing and controlling bioprocess technologies (Dürauer et al. 2024; Puranik et al. 2022; Narayanan et al. 2020). The diverse applications include predictive modeling and process parameters optimization for enhanced product formation, development of virtual sensors for estimation of unmeasured variables, real-time monitoring and control, fault detection and analysis, quality prediction and control in bioprocessing and efficient development of scale-up operations in bioprocesses. Figure 1 depicts the diverse applications of machine learning in bioprocess technologies.

The key players in the biomanufacturing sector are now showing keen interest in AI technologies and directing their investments towards advanced data management techniques which employ machine learning (ML), and the Internet of Things (IoT) (Mondal et al. 2023). The use of ML technologies has gained tremendous attention owing to several reasons. Firstly, ML algorithms have the ability to handle huge amounts of data and can model the complex non-linear

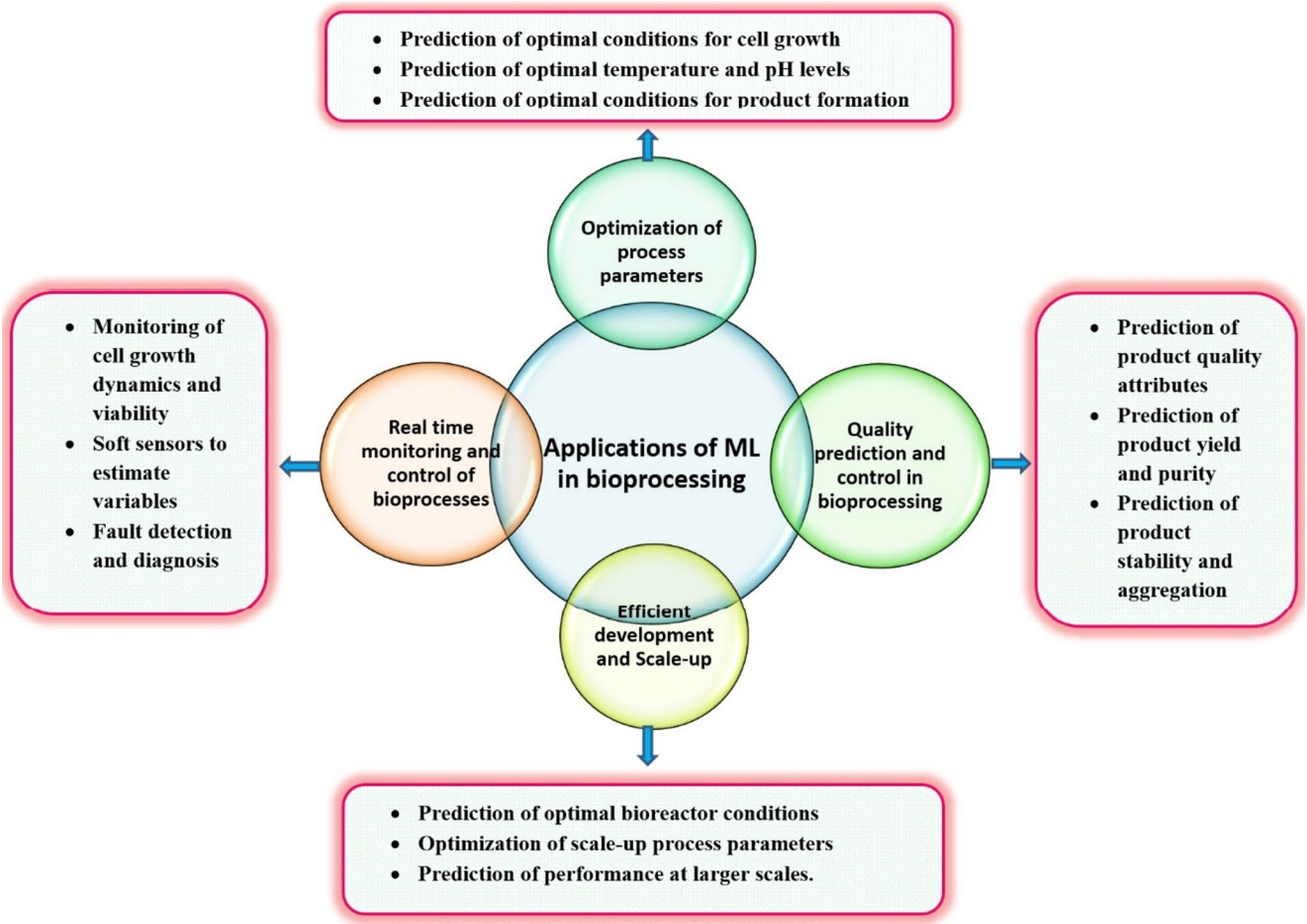


Fig. 1 Applications of machine learning algorithms in bioprocess technologies

behaviour existing between the variables (Nikita et al. 2022). They can be readily integrated into real-time data acquisition systems and can be used to detect deviations, predict potential issues and recommend corrective solutions (Khurshid and Pani 2023; Puranik et al. 2022). ML algorithms can efficiently model scale-up processes by learning from small-scale datasets (Alavijeh et al. 2024). The dynamically adapting nature of ML with the continuously changing environment makes them highly efficient, reliable and sustainable for any industrial process (Helleckes et al. 2023). Different ML algorithms have been majorly applied with high accuracy for enhancing the yield, productivity and detection of diverse metabolites in bioprocesses. Figure 2 depicts the commonly utilized machine learning algorithms for diverse bioprocesses. ML techniques can leverage data from individual unit operations of bioprocesses to develop efficient binary classifiers that can predict whether the end product will meet specified quality thresholds or not (Mowbray et al. 2021). Thus, ML can be judiciously utilized to conserve time and resources and can avoid unnecessary downstream processing operations beforehand. Also, ML-driven automation processes can efficiently choose the suitable analytical methods, accurately forecast system behaviour, and effectively manage uncertainties/imprecise data through robust decision-making abilities (Patel and Shah 2022). Moreover, ML is also a precious tool for reducing uncertainties and risks. By learning from historical data and identifying trends, ML models can help to detect issues before they escalate and can assess the likelihood of success or failure in different scenarios. Thus, understanding and predictions by ML can assist to be more certain of the positive outcomes from a decision.

The application of ML is currently in the early stages of achieving complete automation in the biomanufacturing sector. Still, the last ten years have seen a tremendous surge in promising ML algorithms being utilized in biomanufacturing, optimization and control for classification and regression applications. Classification involves sorting of the data into predefined categories, and various ML algorithms can be trained on historical data to classify new

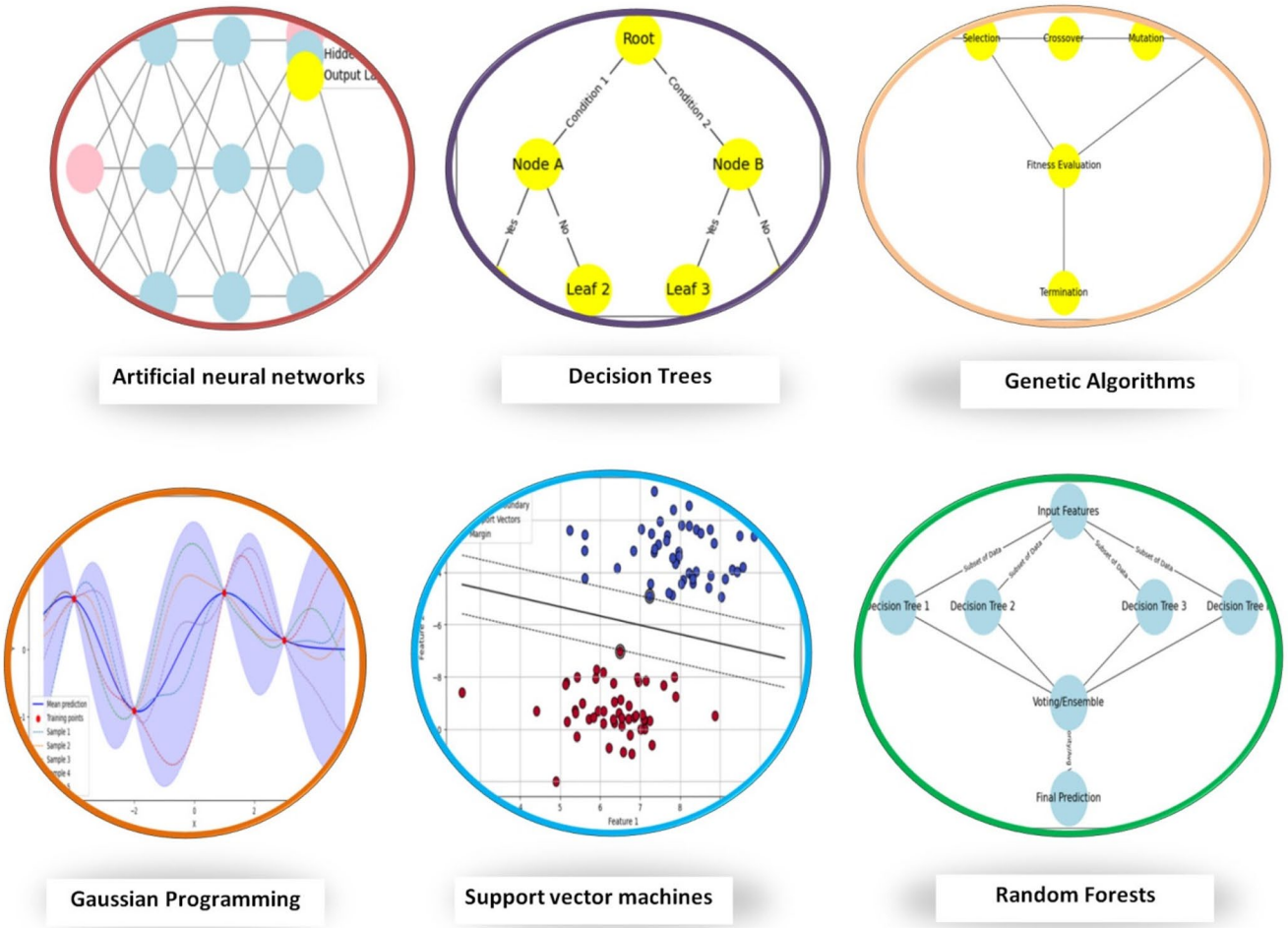


Fig. 2 Commonly utilized machine learning algorithms in diverse bioprocesses

instances accurately. Decision Trees (DT), Random Forests (RF) and Support Vector Machines (SVM) are among the various ML algorithms that can carry out the classification tasks. Regression analysis involves the estimation of the relationship between the dependent and independent variables and to identify the critical variables which are having a significant effect on the target variable. Regression tasks focus on predicting continuous values like yield or biomass concentration in bioprocesses. By analyzing historical data, regression models forecast optimal process conditions, minimize resource usage and thus ensure the appropriate product quality. The regression analysis can be carried out using various ML algorithms viz., Linear regression, Support Vector regression, Neural Networks regression and Gradient Boosting machines. In comparison to conventional methods, ML tools excel in prediction accuracy, adaptability, and scalability. However, ML models come with high development costs, require extensive computational resources, and expert personnel.

The present review is a rigorous and up-to-date approach that starts with the methodology of typical ML workflow and the various ML algorithms available for bioprocess technologies in diverse industries. This is then followed by an in-depth discussion on real-world implementation of ML applications in diverse bioprocessing industries. The major areas covered include the use of ML for enhancing the biofuel and biorefinery bioprocesses, biopharmaceutical bioprocesses, biochemicals production, fermented food and beverages biomanufacturing. The applications of ML algorithms for upstream processing and downstream processing operations are discussed in detail. Lastly, the review provides insights on the patents published by utilizing machine learning algorithms for diverse bioprocessing applications and the future directions for prospective research.

Machine learning fundamentals and its workflow

By the end of the last decade, significant advancements had been made in computer-aided design, architecture, computer vision, and signal processing. Central to these advancements is machine learning (ML), which is a branch of AI. ML enables computers to learn from data, self-improve, and make desirable decisions for each task. The process of model update is carried out by developers or data scientists who manually retrain the model with new data, update the model files and redeploy it. This allows ML systems to adapt and become more accurate over time as they process more data and learn from it (Zhang et al. 2022). High-quality (extensive) datasets are needed to train effective ML models. The datasets should be complete, accurate, sufficiently large, and free from duplicate and irrelevant entries. Moreover, a remarkable feature of machine learning algorithms is their generalization capability, which makes them extremely efficient in future predictions. Generalization capacity is the ability of ML algorithms to predict the results of unseen experiments with high accuracy. This depends on the learning ability of ML algorithms from the training datasets and the application of the gained knowledge to unseen datasets.

Machine learning (ML) of three fundamental types, viz. supervised, unsupervised learning, and semi-supervised learning has been implemented for diverse applications (Patra et al. 2023). Each type serves distinct purposes and is applied to different data scenarios. Algorithms such as SVM, DT, and ANN are frequently used in supervised learning, which deals with labelled data. The utilization of ANN for predicting the yield in diverse bioprocesses is a classic example of supervised learning. Unsupervised learning, on the other hand, deals with the unlabeled data and aims to identify patterns or structures within the dataset (Volk et al. 2020). PCA has been successfully reported for the predictive modelling of viral clearance in a recombinant antibody production process (Panjwani et al. 2021). Clustering is a prevalent unsupervised learning technique which involves grouping of similar data points together based on their inherent properties without predefined categories (Saxena et al. 2017). Apart from clustering, pattern recognition techniques like autoencoders can learn meaningful structure from unlabelled data and find applications in anomaly detection. Also, dimensionality reduction techniques (PCA, t-SNE) are another unsupervised form of learning that finds usage in noise reduction. Semi-supervised learning strikes a balance between both types of learning by utilizing both labelled and unlabelled data. This approach is superior when labelled data is scarce or expensive to obtain.

A typical machine learning workflow involves several key steps designed to build an effective and reliable model. These steps include data collection and preparation, model selection, model training, evaluation, testing, deployment, and continuous monitoring (Bonanni et al. 2023). Figure 3 shows the flowchart of the ML workflow for bioprocess operations. The first step is data collection in which relevant data coming from experiments, manual entries, sensors and databases is gathered. This is followed by data preparation, which is essential for organizing the data. It involves the data pre-processing steps that focus on data cleaning, outlier detection and validation before being fed into the training. One of the major challenges in pre-processing is the detection of outliers in the data. The outliers are anomalous data points that differ significantly from the majority of the data. The presence of outliers can skew machine learning models and lead to unreliable predictions. After detecting the potential outliers, validations are done in order to verify whether those outliers are true outliers (anomalies) or legitimate data points. Further, data cleaning is carried out for the removal

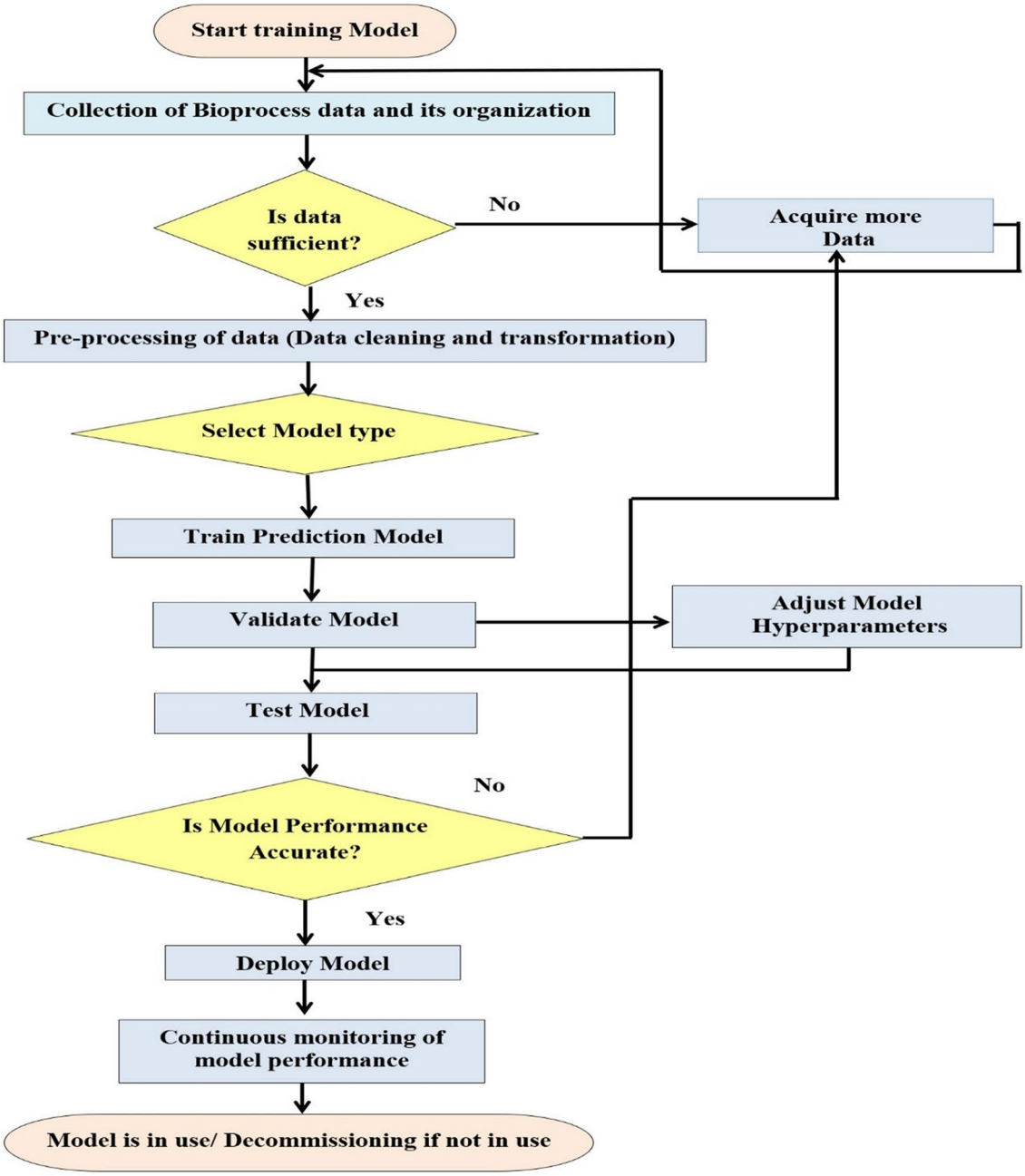


Fig. 3 Flowchart depicting the ML workflow for bioprocess operations

of duplicate data points, handling the missing data and removing the outliers. After cleaning the data, data transformation is carried out by scaling and normalization steps. This step is essential, particularly when the features are measured on different scales. Real-world datasets often contain features with different units and ranges. Without scaling and normalization, features having the larger numerical values can dominate the learning process and lead to biased model performance. The data is then divided into training, validation, and testing sets to allow for unbiased evaluation of the model performance on unseen data in a process called dataset splitting (Lawson et al. 2021). Once the complete data is prepared, model selection follows, which involves choosing the appropriate algorithm for the task. Depending on whether the problem is classification, regression, or clustering, different algorithms are suitable. Tasks like classification involve the use of models that categorize data into predefined classes, and in regression tasks, the purpose is to predict continuous numerical values. For instance, linear regression might be used to predict continuous values, while

decision trees could be used to classify data into categories. The choice of the algorithm significantly influences the ability of the model to learn from the data.

The chosen algorithm is applied to the training data during the model training step. The algorithm learns the relationship between input features and output labels by adjusting its internal parameters. This process involves minimizing a loss function and measures the error between predicted and actual values. Optimization techniques are used to find the best parameters that reduce this error. Afterwards, an evaluation of the model is performed using the validation set. Various metrics are used to assess the model performance, such as accuracy for classification tasks and mean squared error for regression tasks (De Diego et al. 2022; Bedi et al. 2020). This step helps to identify if the model shows overfitting (good performance on training data but poor on new data) or underfitting (poor performance on both training and new data) (Xu et al. 2023). The performance of the constructed model is then evaluated on the test set, which contains data that was not seen during training or validation during the testing phase (Naser and Alavi 2023). This final evaluation provides a realistic measure of the model performance in terms of its ability to generalize to new data and thus ensures its reliability. Continuous monitoring and maintenance are required to ensure that the model remains accurate and relevant over time. Once a model is deployed, it continuously interacts with new and potentially evolving data. Over time, the characteristics of the incoming data change, which can lead to changes in the statistical properties of inputs or changes in the relationship between the inputs and the target. If these changes are not addressed, these shifts can lead to the degradation of the performance of the model. Thus, as new data becomes accessible, the model must be continuously updated to adapt to changing patterns. This ongoing process of monitoring and updating ensures the accuracy, fairness and long-term effectiveness of the model. However, in unsupervised learning, there are no predefined output labels available, and the algorithms search for hidden patterns or groupings within the data. Also, some ML algorithms do not rely on optimization techniques that involve minimizing the loss. Unsupervised machine learning technique like PCA relies on linear algebraic solutions, while in supervised learning, PLS rely on covariance maximization rather than loss minimization.

The choice of a suitable machine learning algorithm depends on a key consideration which screens to identify the number of available data points from experiments and the number of parameters that need to be tuned in the algorithm (Walsh et al. 2022). A data point represents an experiment conducted at a specific time to determine the output variable by providing input variables. A parameter is an internal variable within the model that is estimated based on the data points and can be adjusted. For datasets consisting of a very large number of data points, deep learning approaches that require tuning a large number of parameters are suitable (Alzubaidi et al. 2021). However, this must not be the default choice for every large dataset, as deep learning methods are often time-consuming and computationally expensive to train. The choice of the model should depend on the data type, problem complexity and available resources. Many classical ML algorithms, including the gradient boosting machines, support vector machines, are more appropriate in real-world scenarios where speed, interpretability and simplicity are the key requirements. For datasets with fewer data points, the algorithms that deal with a smaller number of parameters are more appropriate. Decision trees and random forests generally involve fewer parameters to optimize compared to deep learning methods like convolutional neural networks (CNNs). Hence, random forests and decision trees are well-suited for medium-sized datasets.

The most widely utilized ML algorithms used in bioprocessing are the artificial neural networks. In the last 5 years, a significant number of studies on machine learning have utilized ANN alone or in combination with other optimization algorithms (Cruz et al. 2022; Aghbashlo et al. 2021). For the construction of the ANN architecture, MATLAB and PYTHON programming platforms are mostly preferred. As the data generated during the bioprocessing is large, complex and non-linear, the designed neural networks consist of multiple layers. The modern ANN types are implemented as multi-layered perceptrons in which the neurons are interconnected to each other and activated with a non-linear transfer or activation function. The input layer contains the raw data obtained from the sensors or other sources and is generally the process parameters affecting the output of the process. The typical parameters during the bioprocessing are pH, temperature, substrate concentration, oxygen levels and agitation levels. For every input entered in the input layer, appropriate weights and biases are added. Hidden layers are present between the input and output layers of an ANN. The hidden layer consists of more than one layer of neurons and generates a weighted sum of inputs by adding weights and biases to each input. The hidden layer also possesses a non-linear activation function (sigmoid, tanh), which introduces non-linearity into the model. The function of the hidden layer is to capture the non-linear relationships existing between the data variables. The output layer is responsible for the final prediction or classification. Also, various hyperparameters (number of hidden layers, learning rate, activation function, number of epochs) need to be tuned in order to achieve accuracy, generalisation and robustness of the model. The optimization of the hyperparameters in the ANN-specific bioprocess model is generally

carried out using a Bayesian optimization approach and is implemented before the final training phase. An extremely efficient feature of ANN is its generalization capacity, and it determines how well the network can make predictions or decisions in real-world scenarios. A neural network with strong generalization capability can perform accurately on unseen data and thus provides robust and reliable performance across a wide range of input data. Table 1 denotes the various ML algorithms available for performing diverse tasks.

Real-world implementations of ML in bioprocess technologies

Upstream processing is the initial phase in bioprocessing in which the microorganisms and the culture medium are judiciously prepared to initiate the formation of desired metabolites. It plays a key role in the purification, yield and productivity of the final product. Upstream processing starts with the inoculum development stage, which is followed by the preparation and optimization of the culture medium. The culture medium must provide all the necessary nutrients that must favour the optimum growth and yield of the desired product. Different components of the culture medium in the form of carbon and nitrogen sources, vitamins, minerals, precursors and inducers are carefully selected and their optimum concentration is determined in the medium optimization step. Earlier, various statistical techniques such as Plackett-Burman design, one-factor-at-a-time approach, and response surface designs were extensively utilized for performing the medium optimization. Recently, different machine learning techniques have been found to be useful in the optimization of medium components. ANN (Artificial neural networks), GA (Genetic Algorithms), ANN in combination with GA have been significantly reported in different studies (Lau et al. 2023; Amenaghawon et al. 2022; Rabiya et al. 2022). Although statistical techniques are powerful prediction tools yet they struggle to decipher the complex non-linear relationships between the process variables of the culture medium. Moreover, as the number of variables increases, the computational complexity also increases. The statistical methods become cumbersome under high-dimensional data and lead to overfitting of results. Under similar conditions, the ML algorithms dynamically adapt to the changes and optimize the medium composition in real-time conditions. Thus, the majority of the studies focused on optimizing the concentration of the medium components present in the culture medium for improving the product yield or productivity. Also, ML models have been successfully utilized to find out the optimal conditions of physicochemical process parameters such as pH, temperature, humidity, moisture content, salt concentration and many others. Similarly, product recovery is an essential part of any industrial bioprocessing operation. After the broth is extracted from the bioreactor. It is treated in a variety of purification steps to recover the desired product in a process called downstream processing. The ML approach was found to be the best in the recombinant pharmaceutical production process, which involves multiple unit operations during purification. Among the different purification steps, filtration and chromatography are the most widely utilized unit operations. A recent study involving the purification of monoclonal antibodies reported the use of the ANN machine learning model for the prediction of loading capacities of depth filters in commercial production bioprocess. Different factors, including the turbidity, cell counts and viability, were utilized to estimate the changes in the pressure with respect to time across the filter (Agarwal et al. 2016). Another study utilized an ANN hybrid model strategy to determine the flux dynamics and the complete duration of the ultrafiltration unit operation under the varying input process parameters (Krippl et al. 2020a). Fouling is a significant problem that critically hinders the separation in ultrafiltration processes. A study has been carried out to predict the rate of fouling and the filtration flux with changing concentrations of the incoming feed. A hybrid model using a combination of the ML model of ANN and a mechanistic model of stagnant film theory was found to provide promising results in comparison to the mechanistic model alone (Krippl et al. 2020b). Similarly, an integrated process model was developed using a combination of machine learning techniques. Gradient boosting machines/Gaussian programming were utilized in combination with Genetic algorithms to optimize the solubilisation and refolding processes during downstream processing of proteins. A two-fold increase in productivity was observed for the purification of inclusion bodies in the recombinant process (Walther et al. 2022). Another remarkable utilization of ML-based strategy called ELUDE was seen for the purification of peptides (enkephalin derivatives) using RPLC (Reverse phase liquid chromatography). The ELUDE strategy utilized an SVM-based regression model for predicting the retention time of peptides in the column on the basis of the amino acid compositions of peptides. Various promising outcomes were observed in the reported in-silico ML approach, such as the determination of appropriate gradient conditions and a decrease in the number of total experiments during RPLC purification of peptides (Samuelsson et al. 2019).

The comparison of ML approaches has also been utilized and reported for the continuous production of monoclonal antibodies. Five different ML algorithms, namely DNN, RF, DT, SVM regression and GBM regression, were used during the 3-step chromatography purification. The models were compared and selected on the basis of low BIC (Bayesian

Table 1 Diverse ML algorithms available for bioprocess technologies

ML algorithm type / Description	Advantages	References
Artificial Neural Networks (ANN) ANNs are inspired by the human brain structure and consist of numerous interconnected nodes in input, hidden, and output layers. These networks are best suited for modeling complex non-linear relationships, which makes them highly effective in bioprocess industries. ANNs have the ability to model microbial growth and product formation kinetics, can predict equipment failures for proactive maintenance, ensure consistent product quality in pharmaceutical manufacturing, perform classification and can also optimize the bioreactor conditions in real time by combining with different optimization approaches. They can predict the yield of bioproducts like enzymes and biofuels with high accuracy by analyzing historical data.	Efficiently utilized for non-linear regression analysis. Key advantages of ANNs include their ability to model complex relationships between process parameters and efficiently handle large datasets.	Sharma et al. (2024), Konishi (2020), Monroy et al. (2018)
Decision Trees (DT) DTs have the ability to handle both categorical and numerical data. These trees model decisions and their possible outcomes by recursively splitting the dataset into smaller subsets, creating a tree-like structure with nodes and branches. The root node represents the entire dataset and starts the splitting process based on the most significant feature. Internal nodes represent decision points that further divide the data. In contrast, leaf nodes represent the final outcome or prediction, such as a class label in classification tasks or a numerical value in regression tasks. In the bioprocessing industry, decision trees enhance various aspects such as process optimization, quality control, fault diagnosis, predictive maintenance, and yield prediction.	DTs are best suited to identify patterns and predict future yields under varying conditions. The tree structure allows stakeholders to easily grasp how decisions are made, enhancing transparency and regulatory compliance.	Helleckes et al. (2023), Bapat and Wangikar (2004), Coşgun et al. (2022)
Gradient Boosting Machines (GBM) GBMs initially construct a large number of models in multiple stages which are further trained and utilized to create a final model that shows the robust predictive capabilities. The sequential model generation stages are extremely effective in rectifying the errors in previously generated models. They offer accurate modelling and precise control in bioprocesses, which often exhibit highly complex non-linear relationships between process variables.	GBMs can be utilized for both the regression and classification tasks. They are highly useful to overcome overfitting problems that usually occur in individual ML models.	Wang et al. 2024a, b; Mondal et al. (2023)
Random Forests (RF) Random Forests is an ensemble learning ML technique which comprises multiple decision trees, each trained on different subsets of the data, and then aggregating their predictions to enhance performance and reduce overfitting. The final prediction is made by majority voting criteria in classification tasks and by averaging in regression tasks. In the bioprocessing industry, RF are used for process optimization, quality control, fault detection, predictive maintenance, and yield prediction.	High predictive accuracy due to ensemble averaging, robustness to overfitting, ability to provide insights into feature importance, and versatility in handling both classification and regression problems. Fault detection and diagnosis is another critical application.	Pacheco et al. (2023); Helleckes et al. (2023)
Support Vector Machine (SVM) SVMs are supervised ML algorithms that are found to be useful in classification, regression and outlier detection tasks. SVM possess the ability to separate the data into distinct classes based on the hyperplane concept. In the bioprocessing area, SVM finds applications in process monitoring and control, optimizations resulting in yield enhancements, genomics and proteomics data analysis.	SVM can be efficiently utilized for linear as well as non-linear problems. SVM is efficient in handling high-dimensional data by avoiding overfitting.	(Sultana et al. 2022), Zhang et al. (2012)
Gaussian programming/processes Gaussian Processes are probabilistic (non-parametric) models that are best suited for regression tasks in which uncertainty quantification is an essential requirement. The process of making predictions with Gaussian Processes is probabilistic and they can provide a mean prediction along with the confidence interval. For a given set of training data points, the GP model generates predictions for new input data by considering both the mean function and the covariance function. Their ability to integrate data from multiple sources and model complex relationships makes them particularly suited for the variable nature of bioprocessing (Walsh et al. 2022).	Gaussian processes have the ability to model the dynamics of bioprocesses over time as they can efficiently capture the relationships between the process variables and the outputs.	Yeboah et al. (2024), Walsh et al. (2022)
Genetic Algorithms Genetic Algorithms (GAs) are inspired by the principles of natural selection and are designed to solve complex optimization problems. Genes are represented as a combination of binary digits '1' and '0'. A chromosome is composed of these genes. Mixing and matching of the strings is then utilized to find the solutions that result in a superior offspring. This process involves several key steps of selection, crossover, and mutation. However, their effectiveness relies entirely on a separate model (mechanistic or data-driven) that evaluates each solution. Thus, GA relies on a fitness function that serves as an external evaluator to guide the optimization process. In bioprocessing, GAs can be efficiently utilized to maximize yield and productivity by optimizing fermentation parameters.	GAs are particularly advantageous because they perform global searches, reduce the risk of getting trapped in local optima, and can handle nonlinear, complex, and multimodal optimization problems.	Dixit et al. (2023)

Table 1 (continued)

ML algorithm type / Description	Advantages	References
Deep Reinforcement Learning (DRL) It is a type of ML algorithm that can handle highly complex conditions of a bioprocess control environment. The principle involves interactions between an agent and the environment that result in actions aimed at maximising the rewards in the form of bioprocess control. The estimation of reward typically requires a model embedded in the environment. This can be a mechanistic model, a machine learning model or the system itself. Deep Reinforcement learning (DRL) has been extensively utilized in designing and optimizing novel soft controllers.	Extremely useful for the reduction of online computation time in different bioprocess control strategies.	Li et al. (2024)
Principal Component Analysis (PCA) This statistical technique has the ability to simplify complex datasets to transform them into a lower-dimensional form while simultaneously retaining essential information. In bioprocessing, where the datasets are often complex and involve several process variables, PCA can be efficiently used to reduce the complexity of analysis without sacrificing crucial insights. PCA assists in the optimization of bioprocesses by identifying key parameters for attaining maximum productivity with minimum resources and time.	PCA can be extremely useful for understanding data patterns and has been successfully implemented in analyses of biochemical data in diverse fermentation and pharmaceutical processes.	Panjwani et al. (2021); Wang et al. (2023a, b)
Partial Least Squares (PLS) and Partial Least Squares Discriminant Analysis (PLS-DA) These methods are particularly suited for analyzing complex datasets where the number of variables is large and multicollinearity is present. They are widely used in bioprocessing for regression and classification tasks. PLS regression is used to model and predict product concentration, biomass yield, or process efficiency in bioprocesses. PLS-DA modifies the PLS algorithm to handle categorical response variables and thus making it suitable for classification tasks.	These techniques are valued for their interpretability and robustness in handling noisy data, and thus making them highly effective for monitoring, control, and quality assessment in bioprocess environments.	Rathore et al. (2023), Kirdar et al. (2010)

information criterion) values and high R^2. Decision Tree and Random Forest algorithms were found to be the best among all algorithms for real-time measurement and predictions of chromatography purification steps (Saxena et al. 2022). The ML algorithms can efficiently utilize the data from diverse sensors and technologies for real-time monitoring and control. The diverse sensors include the UV-Vis absorption sensors, conductivity sensors, flow rate sensors, refractive index sensors, near-infrared (NIR) spectroscopy and Raman spectroscopy. Moreover, the further generation and integration of genomics, proteomics and metabolomics data will result in the generation of better, more reliable datasets that can accurately measure the process parameters and can lead to a deeper understanding of biological systems and more accurate predictive models, leading to better-optimised bioprocesses. A study reported the successful utilization of a support vector machine ML algorithm in predicting the essential genes for *Escherichia coli* metabolism and survival (Nandi et al. 2017). A study for enhancing the lycopene yield was reported using the ANN-GA ML algorithms. The combinatorial approach resulted in the accurate prediction of target genes responsible for engineered lycopene production in *Deinococcus radiodurans* (Kang et al. 2023).

Machine learning for enhancing biofuel processes

One of the primary aims of the bioprocessing industry is to enhance the production of biofuels and bioenergy products. Considerable research is going towards developing biofuel processes as they are sustainable alternatives to traditional fossil fuels (Manikandan et al. 2022; Ambaye et al. 2021; Mahapatra et al. 2021). Bioethanol, biodiesel, biohydrogen, biogas, and biobutanol are the major biofuels that are getting tremendous attention owing to their renewable nature. Figure 4 depicts the diverse utilities of ML algorithms for enhancing biofuel processes. Table 2 denotes the studies reported on biofuels production using machine learning tools. The biofuel industry and biorefinery concept have experienced significant advancements, owing to the incorporation of various machine learning (ML) models (Okolie 2024; Kavitha et al. 2024; Sharmila et al. 2024). These models were found to be crucial in elucidating and optimizing the intricate nonlinear relationships among various operating parameters, resulting in enhanced biofuel production (Bradford et al. 2018). Apart from enhancing the yield, the application of ML models leads to a reduction in process costs due to the reduction in the number of experiments. The importance of ML models has been reported in a study on biodiesel production using microalgal biomass. Microalgal oil was utilized for the production of biodiesel. ML-based super learning models such as Bayesian optimization algorithm (BOA), ANN, SVM and their different combinations (BOA-ANN and BOA-SVM) were utilized during modeling. A novel BOA-SVM hybrid modeling tool exhibited the optimal performance, resulting in 36.03% yield enhancement (Sultana et al. 2022). The utilization of ML in biofuel production processes can result in economic benefits and a reduction of overall biofuel production costs. The ANN machine learning model was successfully utilized to

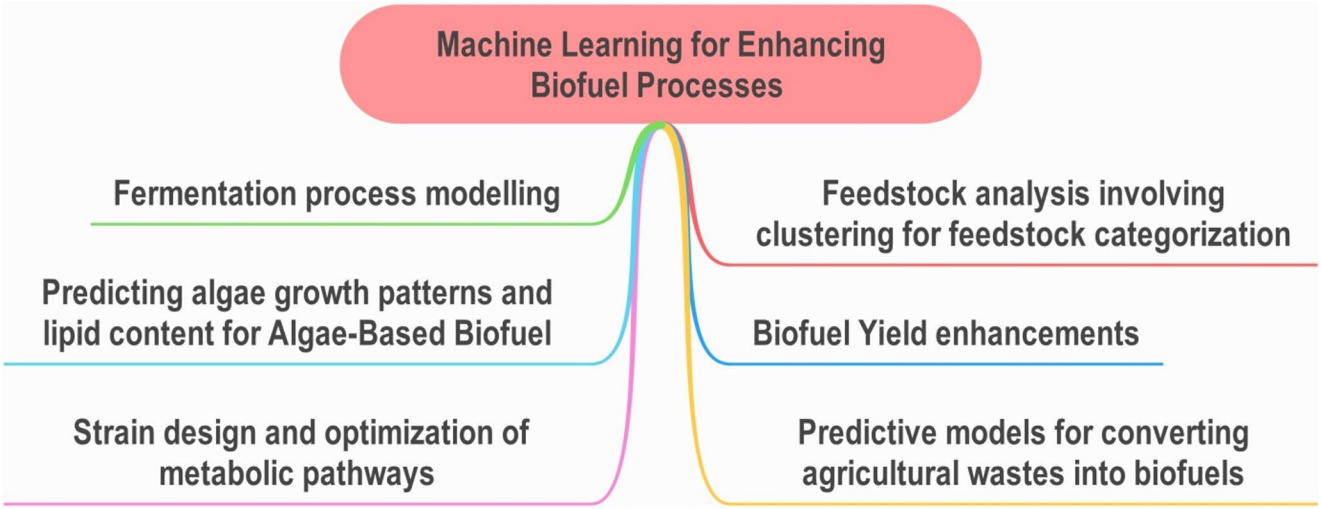


Fig. 4 Diverse utilities of ML algorithms for enhancing biofuel processes

evaluate the economics of biohydrogen production using different biomass feedstocks (Rodgers et al. 2024). A different ML algorithm has been reported to maximize biofuel production and productivity.

In biofuel production processes, a paradigm shift is seen in the usage of the substrates/raw materials for their production. Initially, food-based feedstocks (food crops) were utilized as raw materials, but due to the population explosion and accompanying food shortages, a shift is seen towards the second or third-generation feedstocks (Kumar and Verma 2021; Bardhan et al. 2015). Second-generation feedstocks are derived from non-edible lignocellulosic biomass (agricultural and forestry residues) while third-generation feedstocks on based on algal biomass. One of the major hurdles for utilization of lignocellulosic raw materials (non-food-based feedstocks) or second-generation feedstocks for biofuel production is the intricate presence of lignin polymer with cellulose and hemicellulose fractions (David et al. 2023; Beig et al. 2021). The lignin acts as a barrier during the deconstruction of lignocellulosic. A study reported the usage of the Random Forest ML method as a prediction recovery tool for determining the cellulose-rich fractions in lignocellulosic biomass (Phromphithak et al. 2021).

Currently, a significant number of recent studies have focused on biofuel production from third-generation feedstocks/substrates. The sources of these substrates are the microalgae and macroalgae, which are rich sources of starch and lipids, with faster growth rates and sustainability, as there is no competition with food crops for arable land (Bhatia et al. 2022; Abbasi et al. 2021; Kumar et al. 2020). Various ML approaches have been explored for biofuel development processes using the third-generation feedstocks. In a study carried out on biodiesel production from algae, various process parameters, including methanol solvent to oil percentage, KOH catalyst, and reaction time were optimized to enhance the resulting yield. The statistical methodology of RSM and the ML methodology of ANN were then compared for the determination of biodiesel production. The ANN model outperformed the RSM approach for the conversion of low-temperature-based transesterification of microalgal oil to biofuel (Garg and Jain 2020). Another study reported the use of different ML algorithms for predicting the microalgal bio-oil characteristics using a hydrothermal liquefaction process. Gradient boosting regression performed better than the random forest algorithms, and the obtained results demonstrated the significance of ML-assisted HTL Process for desirable bio-oil production (Zhang et al. 2021). The ML algorithms have also been successfully tested to identify the optimal photobioreactor design and configurations for biofuel production utilizing the microalgae. An innovative strategy involving deep neural networks modeling was employed to reduce the computational time in comparison to conventional CFD and kinetic models. A convolutional neural network predicted the optimal number of holes in the sparger design of a 120 L photobioreactor and also accurately predicted the incident light intensity required for the production of novel bisabolene biofuel from green microalgae (del Rio-Chanona et al. 2019). A combined multi-objective approach using Genetic Algorithms and Fuzzy logic has been successfully utilized to optimize the phycobiliproteins production in *Anabaena* sp. of blue-green algae (Saini et al. 2020). In another study, deep learning techniques of machine learning were adapted to predict the crucial process parameters on the basis of real-time sensor data obtained from anaerobic digestion (AD) processes (Jia et al. 2022). Moreover, image recognition technology combined with Machine learning technologies

Table 2 Machine learning implementations for optimization of biofuel production processes

Substrates	ML algorithms used	Remarks	References
Bioethanol Production			
Ulva prolifera macroalgal biomass	ANN-GA	0.242 g/g ethanol per g of reducing sugar of yield obtained.	Dave et al. (2021)
Agricultural wastes	Gaussian Process Regression (GPR) and Artificial Neural Networks	Accurate prediction of bioethanol yield was found using the squared exponential GPR model.	Yeboah et al. (2024)
Glucose and Corn Steep Liquor	ANN	17.40 g/L of bioethanol obtained.	Taiwo and Musonge (2023)
Corn Cob and Corn stover	Deep neural network	Volatile compositions of different lignocellulosic biomass were obtained and examined for bioethanol production.	Konishi (2020)
Sugar from Sugar mills	Deep learning ANN model	Industrial datasets from sugar mills were utilized to enhance bioethanol production. The influential parameters are cell concentration before fermentation, water hardness and pH.	Niaze et al. (2023)
Sugarcane	ANN and PSO	10% increase in bioethanol concentration	Pereira et al. (2020)
Ricinus communis, *Saccharum officinarum* and *Saccharum spontaneum*	feed-forward ANN	7.86% (v/v) of ethanol concentration was produced under optimized conditions, and an ethanol yield of 62.01 g/L was obtained.	Althuri et al. (2017)
Biohydrogen production			
Wastewater	LightGBM, BagXGBoost, KNN, CatBoost and DT	The CatBoost ML algorithm depicted the optimum performance.	Bandpey et al. (2024)
Sucrose	SVM, XGBoost, DT, RF, kNN	XGBoost resulted in the highest production. Volatile suspended solids, Redox potential, and Butyrate were identified as the most influential factors.	Wang et al. (2024a, b)
Volatile Fatty Acids	Deep learning neural network (DNN)	The DNN model was found to be better than regression models. The models were utilized to predict the effect of time intervals on the production of biogas compounds. The increase in the butyric/acetic acid ratio resulted in enhanced biohydrogen production.	Mahmoodi-Eshkaftaki et al. (2024)
Wastewater	GBM, SVM, RF, AdaBoost	Dark fermentation process for H_2 production. All models showed high R^2 and low MSE values.	Hosseinzadeh et al. (2022)
Corn straw acid hydrolysate	Multi-layered perceptron (MLP), Levenberg-Marquardt ML algorithm, and RSM	The Levenberg-Marquardt algorithm was used for microbial growth kinetics. MLP and RSM were used to determine the electron-equivalent balance (EEB).	Wang et al., (2022)
Tryptone yeast extract- ammonium acetate	ANN	Concomitant production of hydrogen and biobutanol.	Yadav et al. (2021)
Sugarcane Molasses	ANN	84.33 mL of biohydrogen production.	Whiteman and Kana (2014)
Short-chain fatty acids	ANN	ANN models were used to predict the hydrogen yield, rate of production and accumulated hydrogen.	Sydney et al. (2020)
Chlorella vulgaris biomass grown on BG11 media	ANN	Oleic acid concentration of 745.21 mg/L was obtained as a product.	Liyanaarachchi et al. (2021)
Biomethane production			
Cellulose	GBM, ANN, RF, SVM	Hybrid modelling was performed to examine the biomethane potential of biomass. The regression model generated using Gradient Boosting was found to be accurate.	Adeleke et al. (2024)
Ligno-cellulosic Biomass	K-nearest neighbor	Biomethane potential was evaluated on diverse lignocellulosics. The amount of lignin present has a direct effect on methane production.	Wang et al. (2023a, b)
Mixed substrates	Elasticnet, Random forest, and extreme gradient boosting	Industrial-scale methane production was studied on 4-year dataset obtained from an industrial-scale co-digestion (ACoD) facility. XGBoost outperformed the other ML algorithms	De Clercq et al. (2022)
Biodiesel production			
Algal Biomass	Decision Trees	The critical factors for lipid production enhancement were glucose and pH. C/N content and production duration affected the biomass productivity.	Coşgun et al. (2022)

Table 2 (continued)

Substrates	ML algorithms used	Remarks	References
Microalgal biomass	Decision Trees and Associative Rule Mining	The ML algorithm resulted in the identification of 11 different blends of parameters for enhanced biomass productivity and 13 different blends for enhancing lipid productivity.	Coşgun et al. (2021)
Palm oil	MLP, kNN, ANN, GPR	ANN modeling was found to be superior in comparison to other algorithms in optimization for FAME production.	Jin et al. (2022)
Glucose, Corn steep liquor. Yeast extract added to basal medium	ANN, SVM and GA	Synergistic relationships between biomass, concentration of lipids and rate of conversion was revealed using *Rhodotorula glutinis* as producer organism.	Zhang et al. (2024)

has been employed for the quick detection of the maturity of compost formation (Wang et al. 2024a, b).

In a biorefinery concept where the focus is on converting biomass into biofuels and biochemicals, machine learning techniques are also being utilized to develop predictive models of enzymatic hydrolysis of substrates and fermentation processes. The experimental data on substrate composition, enzyme kinetics, growth kinetics, substrate uptake rate, product formation rate and microbial metabolism at different process parameters (pH, temperature, agitation rate, aeration rate, inoculum size, fermentation time, dissolved oxygen concentration) are assimilated, and then utilized to construct dynamic process models using diverse ML algorithms. These models facilitate real-time simulations of different process scenarios and allow operators to assess and optimize the variations in parameters for improving product yields and process efficiency. Artificial neural networks (ANNs) have been significantly explored in biohydrogen production owing to their capability to capture complex, nonlinear relationships between operational and monitoring variables. For instance, Nikhil et al. (2008) developed an ANN model to forecast the biohydrogen production rates using input variables such as hydraulic retention time (HRT), biomass concentration, pH, and recycle ratio in a carbohydrate-based bioreactor system. The model successfully predicted the hydrogen production rate as output and thus offers a robust control framework for pilot-scale operations. However, the model was limited to predicting a single output at a time. To address multi-output systems, Wang and Wan (2009) developed a feed-forward ANN model to predict both biohydrogen yield and substrate consumption by mixed cultures. Their model exhibited high accuracy with R^2 values of 0.99 and 0.98 for hydrogen production and substrate consumption, respectively.

Recently, multi-objective hybrid models were reported for biohydrogen production. Wang et al. (2021a, b) employed a hybrid model incorporating ANN (machine learning approach) with response Surface Methodology (RSM) (statistical approach). The hybrid model effectively optimized the crucial process parameters including the microbial dose levels, carbon sources, metal cofactor requirements, and pH as input variables and proving to be assertive and decisive for enhancing biohydrogen production. Similarly, Pereira et al. (2020) integrated a Particle Swarm Optimization (PSO) algorithm with an ANN model to predict bioethanol production rates. The ANN-PSO combination optimized the process parameters to maximize ethanol production with high accuracy (R^2 values ~ 0.9).

Machine learning for enhancing biochemical production

The bioprocessing industry has made a remarkable contribution to human society by facilitating the industrial production of a diverse variety of enzymes, industrial biopolymers and solvents (Sodhi et al. 2022; Usmani et al. 2021; Srivastava et al. 2020). Yet, many of the bioprocesses have low yield and productivity. Machine learning methodologies are therefore employed to optimize the medium composition, nutrient supplementation, physico-chemical process parameters and aeration rates during the production (Kalwani et al. 2024; Li et al. 2021). Figure 5 depicts the diverse applications of ML algorithms for enhancing biochemical production processes. ML algorithms aim to identify the optimal operating conditions, resulting in maximization of the product yield and minimization of the resource consumption (Katongtung et al. 2022; Asrul et al. 2022; del Rio-Chanona et al. 2016). These ML technologies are also employed for real-time applications where these algorithms dynamically adjust process variables to maintain optimal conditions, thus enhancing overall process efficiency and productivity (Ma et al. 2020). In a recent study on bioprocessing for biopolymers, polyhydroxybutyrate (PHB) production from *Cupriavidus necator* was predicted using multiple ML algorithms. Raw spectra from visible and near-infrared spectroscopy were used to construct ML models. The partial least squares ML model accurately predicted the highest PHB production from the alkaline medium extracted from corn stover (Li et al. 2020). An interesting approach of machine learning towards the enhanced production of polyhydroxyalkanoates is seen in a study in which a combination of mechanistic modelling and deep learning modelling was utilized for

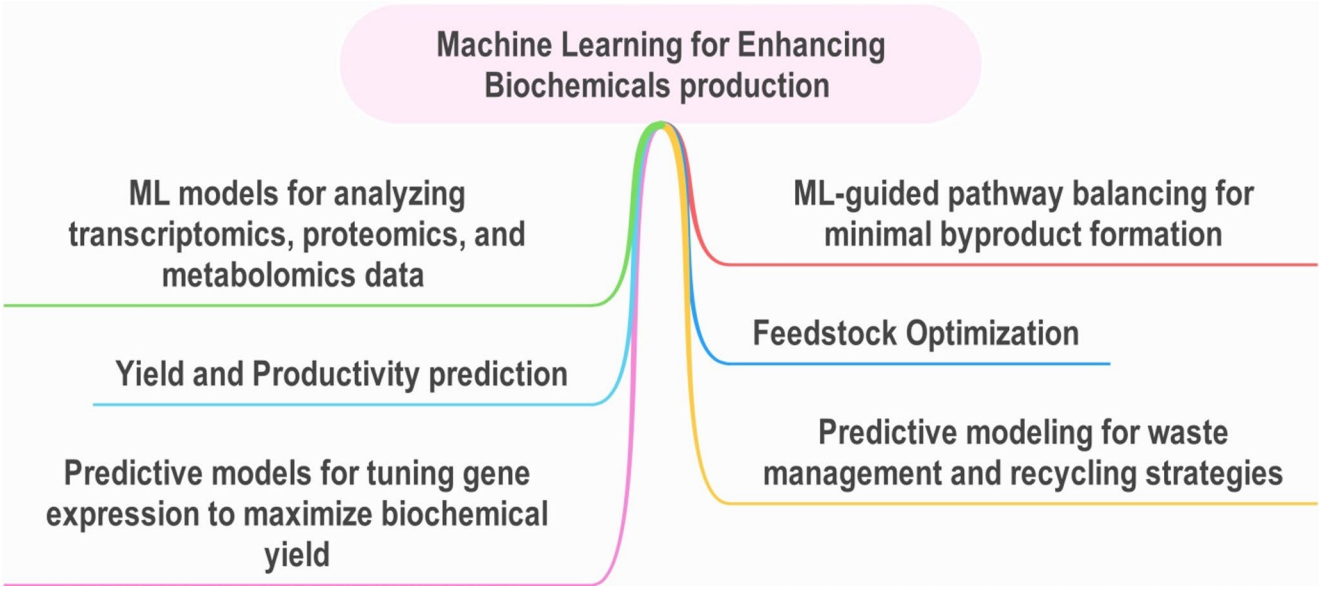


Fig. 5 Applications of ML algorithms for enhancing biochemicals production processes

the enrichment of PHA-accumulating bacteria. Recurrent neural networks (RNNs) accurately predicted the outcomes with an R^2 value of 0.991 (Xu et al. 2022). In addition to polymers, the bioprocessing industry produces diverse kinds of enzymes for numerous applications. These include industrial enzymes, therapeutic enzymes, and enzymes for biorefinery and food applications. A large number of ML studies reported on biochemical production aim to predict and enhance the enzyme yields using regression-based optimization studies (Table 3). ML technologies have also been effectively utilized for the directed evolution studies on enzymes (Wittmann et al. 2021; Buller et al. 2021; Saito et al. 2021). Directed evolution involves numerous rounds of mutagenesis and selection experiments that aim to evolve the proteins to achieve desired properties. In an investigation to examine the significance of ML in directed evolution, sortase A enzyme variants were examined using a Gaussian process ML algorithm (implemented in Bayesian optimization software COMBO) to explore the effects of variation that showed the highest enzyme activity (Oikawa et al.,2021). Similarly, another interesting study reported the generation of an ML model based on partial least squares regression from the amino acid sequences and the fitness score of the mutant variants. The model was then subsequently utilized in enhancing the enantioselective property of epoxide hydrolase enzyme of *Aspergillus niger* (Cadet et al. 2018). ML-directed evolution has been successfully reported as a promising enzyme engineering approach for the construction of a library containing mutants with high activity (Ma et al. 2021).

Machine learning for enhancing biopharmaceuticals production

The critical importance of machine learning in pharmaceutical bioprocessing started with studies that aimed to predict and optimize the yields of biopharmaceutical compounds produced by microbial bioprocessing. Table 4 reports the various studies in which the product yield of biopharmaceuticals has been enhanced using different ML algorithms. The active biological products include a diverse range of antibiotics, immunosuppressants, vaccines and monoclonal antibodies. Besides improving the overall yield, machine learning also finds usage in diverse applications of pharmaceutical industry including, novel anti-microbials drug discovery, protein engineering, identification of potential biomarkers, reduction of variability in the overall process, product quality control, and identification of faults during purification procedures (Rolff et al. 2024; Wang et al. 2023a, b; Narayanan et al. 2021; Xie et al. 2021). Figure 6 depicts the diverse applications of ML algorithms for enhancing biopharmaceutical production processes.

Multivariate Data Analysis (MVDA) algorithms are the most widely adopted class of ML tools within the biopharmaceutical industry due to their interpretability, robustness, and regulatory friendliness. These include the principal component analysis (PCA), partial least squares (PLS) regression, and multiple linear regression (MLR) (Rathore et al. 2023). MVDA allows for the identification of critical process parameters (CPPs) and their relationships to critical quality attributes (CQAs). MVDA algorithms are mainly used for process parameters optimization in biopharmaceutical manufacturing (Rathore et al. 2012; Kirdar et al. 2010).

Table 3 Machine learning implementations for optimization of biochemical production processes

Substrates	ML algorithms used	Remarks	References
Biopolymer production			
Sucrose and Urea	GA-ANN	Nitrogen and Phosphorus concentrations were optimized to enhance the polyhydroxyalkanoates (PHA) production. The GA-ANN model with six neurons in the hidden layer resulted in 2.69 g/L of PHA production.	Lhamo et al. (2023)
Sunflower oil, Sucrose	ANN	ANN model resulted in 10.411 g/L PHA production from a sucrose-sunflower oil ratio maintained at 1.729 (w/v). Transition from sucrose to sunflower oil causes an enhancement in PHA production.	Laurence et al. (2024)
Enzymes production			
Wheat straw, rice husk and Bagasse	Bayesian regularization neural network, SVM, Radial basis function neural network, Gaussian process learner	A threefold enhancement in Cellulase production was obtained using *Aspergillus flavus*. Radial basis function neural network was found to be the best ML algorithm in the study.	Singhal et al. (2022)
Peach palm waste	ANN-GA	Yield enhancement of 31.58-fold was observed. The produced carboxymethylcellulase was applied for fibre degradation and saccharification of waste.	Bezerra et al. (2022)
De-oiled rice bran	Multi-objective artificial neural network-genetic algorithm	Enhanced xylanase and cellulase production from rice bran under ML optimized conditions of 1:1 ratio of substrate/moisture content, 3.5 days of incubation and inoculum size of $3*10^6$ spores/mL.	Saharan et al. (2023)
Waste cooking oil	ANN	The ANN model outperformed the RSM approach. Olive oil, KH_2PO_4, and $MgSO_4$ acted as stimulants and stimulated the lipase production.	Amenaghawon et al. (2022)
Cocoa shells	ANN-GA	Physical process parameters were optimized for lipase production in SSF from *Penicillium roqueforti*. 267.7% enhancement of lipase production using ANN-GA compared to classical approach.	de Menezes et al. (2021)
Wheat Bran and Eucalyptus leaves	Genetic Algorithms	Multi-enzyme (Lipase, Endoglucanase and Amylase) production studies were optimized using a multi-objective GA approach. Cost-effective microbial production and its application in deinking are reported.	Dixit et al. (2023)
Dairy effluent	Genetic Algorithm coupled with response surface methodology	150-fold enhanced *esterase* production was obtained using machine learning optimization. Almond oil and casein were found to be the significant components.	Sounderarajan et al. (2024)

Machine learning has been successfully applied to accelerate antimicrobial discovery and development (Zoffmann et al. 2019; Serafim et al. 2020; Stokes et al. 2020; Liu and Stokes 2022). An ML study has been carried out to screen the novel anti-microbial drugs against multidrug-resistant bacterial strains. ANN and decision trees deployment resulted in improvement in hit rates and a faster transition from the pre-clinical to clinical stages of drug development (Durrant and Amaro 2015). Similarly, ML has also been successfully implemented to recognize novel antibiotics. A novel bacterial phenotype-based classification tool was reported to identify the mode of action of new antibiotics. Random forest ML algorithm was employed, and training datasets were prepared on different reference compounds which give distinct and clear fingerprints. The generated model provided the probability of the similarity between the unknown test compounds and those compounds used during the training (Zoffmann et al. 2019). Similarly, monoclonal antibodies (mAbs) are a cornerstone of modern therapeutics as they offer targeted and safe treatments against a wide range of diseases. Mabs are high-value biologics whose market value is expected to increase from US$185.5 billion in 2021 to US$494 billion by 2030 at a CAGR of 11.3% (Khuat et al. 2024). Their production involves highly sophisticated and tightly controlled steps to establish high quality and safety. The downstream processing steps involve multiple chromatography separations that play a critical role in the final yield and purity. In a recent study, deep learning-based convolutional neural network modelling was utilized for predicting the Protein A and Cation exchange chromatography elution concentrations in real-time conditions. Further, the purification was optimized using multi-objective-based Bayesian ML algorithm, resulting in a stable yield obtained during real-time conditions (Alam et al. 2024). Another ML study reported a time-dependent ML pipeline that can precisely predict the most critical parameters and the overall performance of the process during biologics production. The ML approach can classify the whole production process into poor, moderate and high-performing processes within a week (sixth day), even though the harvesting of the culture takes place after two weeks (Gangadharan et al. 2021). Also, a machine learning approach has been utilized

Table 4 Machine learning implementations for optimization of biopharmaceuticals production processes

Substrates	ML algorithms used	Remarks	References
Ground-nut de-oiled cake, soybean and corn gluten meal mixture	ANN	A ternary mixture of agro-substrates was reported to be better than a single substrate alone. ANN was utilized for pH, moisture content and autoclaving period optimization for enhanced L-asparaginase production.	Sharma and Mishra (2024)
Sucrose, Peptone, Uric Acid	ANN GA	*Novel* uricase preparation from isolated *Pseudomonas mosselii* was reported. ANN GA optimization resulted in a 2.02-fold enhancement in enzyme production.	Dudala et al. (2024)
Waste frying oil	ANN	Glycosylated peptide produced from *Pseudomonas aeruginosa* depicted antibacterial, anti-biofilm and anticancer activities against cervical and breast cancer cell lines. Optimization for enhanced production revealed 32 °C temperature, pH 7.6, agitation of 130 rpm, and 66 h of production time as optimum parameters.	Ekpenyong et al. (2021)
Glucose	ANN	Antimicrobial bacteriocin was produced and tested against food-spoiling *Staphylococcus aureus*. ANN and RSM models were applied for optimizing the production process.	Lahiri et al. (2021)
Agro-waste Niger de-oiled cake	ANN	A novel low C/N content substrate was reported for anti-leukemic enzyme production. ANN modeling was performed for the optimization of the process parameters (pH, temperature, moisture content and autoclaving time) involved in the bioprocess.	Sharma and Mishra (2022)
Mannose, Soybean meal	ANN-GA	Immunosuppressant production and optimization using ANN-GA resulted in enhanced titres. L-Lysine acted as a precursor.	Sinha et al. (2014)
Dextrose, Soy flour	GA, DT	GA predicted eleven unique media combinations that resulted in high productivity of Rifamycin B. The DT were utilized to predict media-media interactions. The outcomes showed more than 600% enhancement in productivity.	Bapat and Wangikar (2004)

for the quality control of biopharmaceuticals. A study carried out by researchers reported the applicability of a non-linear KNN model for monitoring the monoclonal antibody product quality in real-time. The system used hundreds of calibration points from Raman spectroscopy to build non-linear ML models that are capable of predicting antibody concentration and aggregate levels every 9.3 s during purification (Wang et al. 2023a, b).

The ML approaches have also been successfully tested along with the mechanistic models in a relatively new approach called hybrid modeling. Hybrid models combine the strengths of mechanistic (first-principle) models and data-driven (machine learning) models. The fusion allows the model to maintain interpretability (from mechanistic models) while also improving accuracy and adaptability (from data-driven models). The hybrid mechanistic/ML strategy reduces the number of experiments to be performed and also leads to increased accuracy due to the training on previous data sets (Walsh et al. 2022). Hybrid models were successfully implemented in predicting the N-linked glycosylation pattern variation of two fusion proteins (Kotidis and Kontoravdi 2020). Similarly, a hybrid ANN/ material balance model was successfully used to determine the cell growth and titre for fed-batch production of therapeutic proteins in mammalian cultures (Narayanan et al. 2019). A bioreactor scale-up study was also carried out to assess the applicability of different ML algorithms in a monoclonal antibody production process (Alavijeh et al. 2024). The ANN model was utilised to assess viable cell density (VCD), a parameter directly linked to nutrient consumption, growth and product formation and the CatBoost algorithm was utilised to predict P/V (Power/Volume) parameter as a scaling criterion.

Although the integration of ML in the biopharmaceutical processes is rapidly transforming drug discovery, clinical development, and biomanufacturing, this innovation needs to be tightly coupled with regulatory oversight in order to ensure end-user safety, data integrity, and ethical compliance. Various regulatory agencies throughout the globe, including the U.S. Food and Drug Administration (FDA), European Medicines Agency (EMA), and World Health organization (WHO) have started adapting the existing frameworks or creating new ones to address the unique challenges and opportunities posed by the AI (Mirakhori and Niazi 2025). The US FDA has issued several guiding documents regarding the use of ML in pharmaceutical applications and is actively developing Good Machine Learning Practice (GMLP) principles in collaboration with international agencies. The FDA requires that the use of ML models that directly impact patient care or drug evaluation must adhere to rigorous standards of validation, transparency, and post-market monitoring (Vora et al. 2023). The European

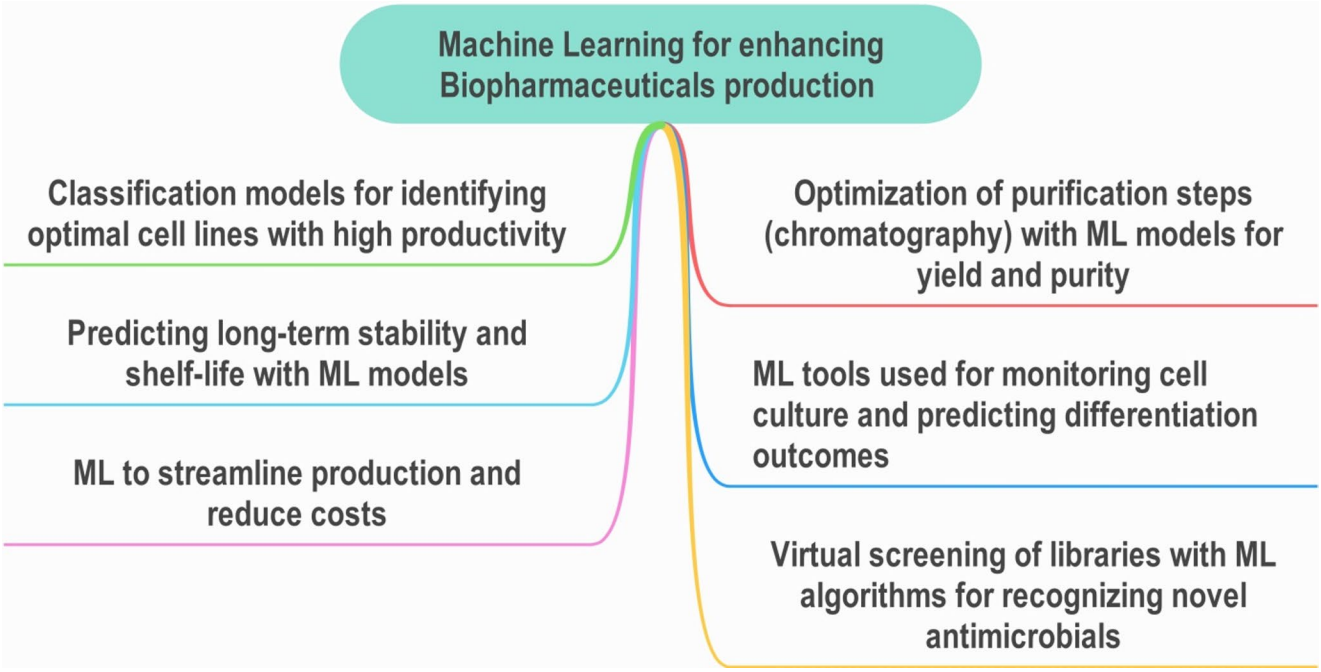


Fig. 6 Applications of ML algorithms for enhancing biopharmaceutical production processes

EMA approach classifies ML models into risk categories, with pharmaceutical applications typically falling under the high-risk category due to their impact on human health and safety. These model systems must comply with the stringent requirements of transparency, human oversight, and risk management. According to the guidelines issued by the WHO on biologics production and their clinical trials, the manufacturing industries must focus on ethical considerations, handling biases, ensuring data privacy and critical validation in order to guarantee that AI incorporation is safe, reliable and meets the efficacy standards. The utilization of AI should comply with good manufacturing practices (GMP) and must enhance the quality of the desired product (Mirakhori and Niazi 2025). Moreover, all aspects of ML development, starting from data sourcing and pre-processing to model training and validation, should be recorded in a manner that will support audits and inspections. The continuous monitoring of the models after deployment, combined with a robust governance framework, will ensure that ML models remain safe and effective over time.

Machine learning in food bioprocessing

The relationship between the food industry and bioprocessing is ancient and dates back to the production of valuable fermentation bioproducts, including vinegar, wine, cheese, beer, yoghurt, sauerkraut, tempeh, probiotics and many others (Sun et al. 2022). Moreover, the fermentation technology also leads to the production of antimicrobial peptides like nisin and natamycin that find widespread usage in food preservation (Kamal et al. 2023). The advent of machine learning (ML) practices has revolutionized the field of food bioprocessing by offering innovative solutions to improve food production, quality, safety and sustainability (Kyaw et al. 2024; Ding et al. 2023). Figure 7 denotes the diverse applications of ML algorithms in food bioprocessing applications. One of the crucial aspects of bioprocessing in the food sector is to enhance food production by ascertaining the optimized conditions. Various ML algorithms have been deployed and reported for enhancing the product yield. Table 5 reports the various studies in which the product yield has been enhanced using different ML algorithms. Apart from enhancing the yield, ML has been significantly applied in enhancing food/beverage quality and safety (Shahrajabian and Sun 2024; Singh et al. 2024; Bowler et al. 2021). Food quality ensures the maintenance of key attributes like texture, flavour, composition, and purity (Feng et al. 2024; Lu et al. 2020). Many foods and beverages are prepared through fermentation routes and involve major bioprocessing steps. A large number of ML studies on food bioprocessing have been carried out on maintaining the essential properties of beer and wine solvents (Hlangwani et al. 2021; Gonzalez Viejo et al. 2019). A study on beer production reportedly applied deep neural network ML methods to improve beer production. The study utilized the previously collected data from ultrasonic sensors for density measurements as the training datasets and then applied the deep neural networks to optimize the beer production monitoring process at an industrial-scale (Bowler et al. 2021). In another study carried out on beer production, ANN and

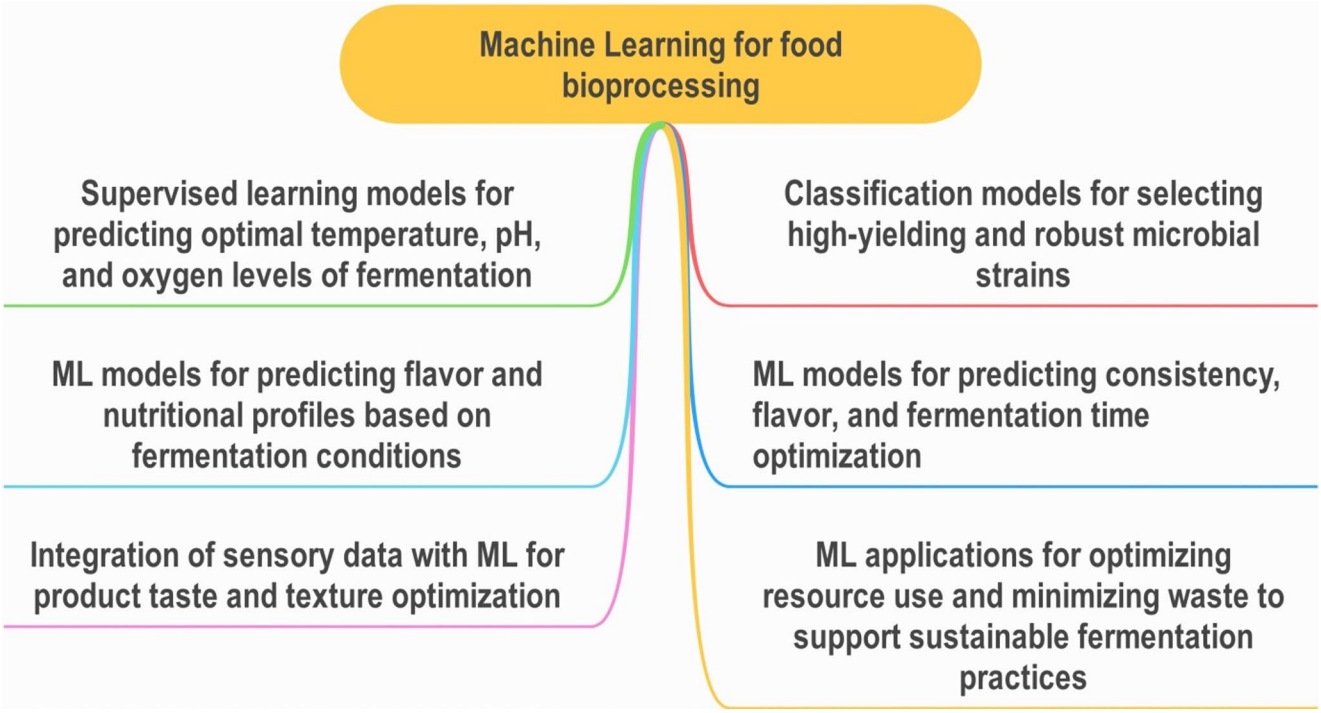


Fig. 7 Diverse applications of ML algorithms in food bioprocessing applications

Table 5 Machine learning implementations for optimization in food bioprocessing

Substrates / Microbial Product	ML algorithms used	Remarks	References
Blueberry juice	ANN-GA	Fermentation process parameters were optimized using ANN-GA modeling for exopolysaccharide production by *Pediococcus pentosaceus*. The model was constructed from the dataset obtained from Visible-Near Infrared spectroscopy.	Yang et al. (2024)
Pineapple waste	PSO, GA	Optimization using two evolutionary ML algorithms resulted in an 11.9% increase in xanthan gum production. KH_2PO_4 and NH_4NO_3 were the two most crucial factors that acted as stimulants.	Amenaghawon et al. (2024)
Serum Culture medium	ANN (Radial basis function)-GA	The cultivated meat production process was optimized for reduced serum requirements. The study identified media components that resulted in a sustainable, low-cost production of cultivated meat.	Nikkhah et al. (2023)
Waste feather and waste peanut oil	ANN-PSO	Prodigiosin production and optimization were performed using ANN-PSO-based ML methodology. The pigment was then screened for antimicrobial, anti-cancer and dyeing of textile fabrics.	Asitok et al. (2023)
Sucrose, NaCl, $NaNO_3$, KH_2PO_4	Hybrid SVM (h-SVM)	Exopolysaccharide (EPS) production process from *Bacillus licheniformis* was optimized using an h-SVM algorithm. A 5-fold enhancement in production yield was observed.	Rabiya and Sen (2022)
Sorghum and Maize	ANN	ANN model was used as a non-linear optimization tool for Umqombothi (a South African traditional beer) bioprocessing development.	Hlangwani et al. (2021)
Sucrose	ANN and fuzzy logic	Glucan and fructan exopolysaccharides were produced from *Leuconostoc mesenteroides.* Hybrid modeling using ANN and fuzzy logic were employed for optimization of the fermentation process.	Kabli et al. (2020)

SVM ML algorithms were utilized for the prediction of acetic acid during industrial-scale beer production processes. The acetic acid content beyond a threshold will impart undesirable taste characteristics to the beer. Therefore, controlling the acetic acid production during the beer fermentation process is critically important to maintain the quality of the beer. The dataset of 146 beer production batches was used to construct ML models. The SVM model performed better than the ANN model in predicting the final output of acetic acid detection (Zhang et al. 2012).

Recently, ML algorithms have been applied in a black tea fermentation study to classify the fermentation process

into three different categories- under-fermented, normal fermented and over-fermented tea varieties. Support vector machines were found to be the best for the accurate detection of fermentation levels (Bhargava et al. 2023). ML algorithms have also been successfully utilized for the identification of shrimp waste origin in different regions in China by utilizing flavour omics datasets. The datasets were constructed on volatile compounds associated with aroma and were measured using GC-MS analysis. Out of 7 different ML algorithms applied, RF, SVM, and ANN were able to predict the geographical origin of shrimp waste with high accuracy (Lu et al. 2022). Similarly, ML approaches have also been applied to cultured meat production. The cultured meat product has already gained significant attention as it can act as a sustainable alternative to prevent the slaughtering of animals and can fulfil the high demands of the future. In a recent study, PCA ML analysis and heat maps were applied to determine whether a correlation exists between the selected morphological parameters and the quality of muscle (porcine) stem cells used for cultured meat production (Wang et al. 2023a, b). Thus, machine learning plays a critical role in food bioprocessing by enhancing their production, taste, texture, flavour and purity.

Machine learning in the monitoring and control of bioprocesses

Real-time monitoring and control of the process variables are extremely critical for optimizing the process parameters and the environmental conditions in bioprocessing (Mowbray et al. 2021; Wasalathanthri et al. 2020). They are required to ensure the efficiency, productivity and safety of the different bioprocess operations by dynamically adjusting to real-time operating conditions. Figure 8 depicts the diverse applications of ML algorithms in the monitoring and control of bioprocess technologies. In modern bioprocessing industries, significant research is going on towards the continuous production of metabolites, including biofuels, enzymes, biopharmaceuticals, biopolymers and many others (Xie 2022; Woodley 2020; Chen et al. 2018). These continuous processes heavily rely on real-time monitoring and control to maintain optimum production conditions, desired efficiency and productivity. Conventionally, monitoring and control operations were carried out using hardware sensors which can monitor pH, temperature, pressure, foam, viscosity, biomass concentration, oxygen and carbon dioxide concentrations, but these sensors are unable to provide information about the specific growth rates and other complex parameters. With the advent of machine learning modelling combined with the mechanistic modelling approaches, soft sensors came into existence. The retrieval of signals from the different bioprocesses is often complex and noisy, and can give rise to offset values due to non-reliable calibration models (Narayanan et al. 2020). The soft sensor structure has a multi-layered structure which relies on multiple calibration models to generate the output. These soft sensors have shown extremely promising outcomes in monitoring and control operations by measuring the process variables which cannot be estimated directly and are impractical or expensive using hardware sensors (Ching et al. 2021; Zhu

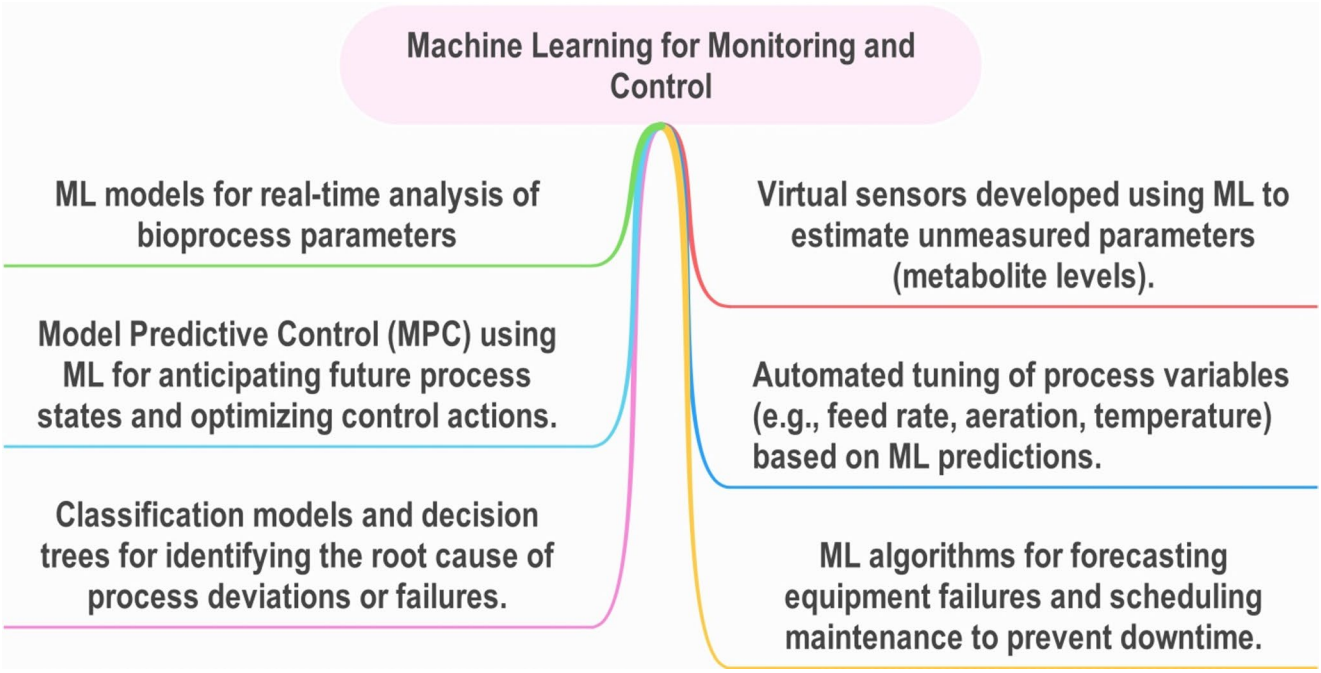


Fig. 8 Applications of ML algorithms in monitoring and control of bioprocess technologies

et al. 2020). Soft sensors involve the generation of sensitive calibration models using different machine learning and mechanistic approaches. They utilize the data using easily measurable variables to generate the models for complex measurements. A variety of ML algorithms like neural networks, gradient boosting, deep neural networks, and support vector machines find usage in the development of soft sensors (Perera et al. 2023; Ching et al. 2022). The algorithms can be easily implemented using MATLAB, Python and R tools to generate prediction models. The reliability of the models was then evaluated in terms of mean squared error (MSE), root mean squared error (RMSE), Mean absolute error (MAE), coefficient of determination and many others. Lastly, the soft sensors need to be regularly tuned with time as a means to maintain the credibility of the predictions on the unseen dataset of the model. Various ML approaches have been efficiently utilized and reported for the development of soft sensors in bioprocessing operations. A deep learning based soft sensor has been reported for streptokinase and penicillin production processes using deep neural networks (DNN). The semi-supervised learning approach of deep learning was found to be the best for processes involving complex biochemical reactions (Gopakumar et al. 2018). In a recent study, various soft sensors were developed in the lysine production process to estimate the concentration of lysine as a product, microbial growth and sucrose as a substrate in real-time fermenter conditions (Tokuyama et al. 2021). In a different study, different ML algorithms were used in different regression problems. In a study carried out on penicillin production, a combination of ANN, fuzzy rule systems and mass balance standard equations was applied to create a hybrid model and a significant enhancement in penicillin production was observed (Galvanauskas et al. 2004). Similarly, in a recent study on the fed-batch process of penicillin antibiotic, several soft sensors were developed using ML algorithms for the accurate prediction of industrial-scale penicillin production (Acosta-Pavas et al. 2024). Anomaly detection is a critical component for monitoring and control systems and is applied in different bioprocess studies (Nguyen et al. 2025; Zeberli et al. 2021). In a recent study on contamination detection during fermentation, machine learning using support vector machine demonstrated excellent accuracy in contamination detection and was also found to be suitable for real-time monitoring systems (Nguyen et al. 2025). A data-driven machine learning approach of anomaly detection and diagnosis was applied to a decontamination study using hydrogen peroxide (vaporous) in the biomanufacturing of a pharmaceutical product. Principal component analysis (PCA) was applied to a production history data of four years obtained from sensor measurements. Different failure boundaries were constructed for different types of anomalies, and a binary principal components detection approach led to the reduction of downtime and an enhancement in production efficiency (Zeberli et al. 2021).

Patents in bioprocessing

Besides the publication of research articles, several patents have also been reported for the applications of ML in bioprocess development. As the field of machine learning is relatively new and its applications in bioprocess development are rare, the last decade has seen an increase in the number of patents granted in bioprocessing. The majority involved the utilization of machine learning for monitoring the real-time process conditions, prediction of the process parameters, including the metabolic state of microorganisms and for bioprocessing applications. Another area in which ML innovations have been reported is the mammalian cell culture, where the key focus is the development of a new cell line that can exhibit enhanced productivity. Table 6 lists the various patents that found usage of machine learning for bioprocessing.

Ethical considerations and challenges

The decision-making driven by machine learning holds great potential in bioprocess optimization and improving product consistency. However, their implementations are also accompanied by several ethical issues. The decision-making systems often work as “black box” models, which results in a lack of explainability and transparency. The ML algorithms in therapeutic bioprocess development rely heavily on data, including proprietary formulations, strain information and operational parameters. Ethical risks can arise due to data breaches or the uninformed use of data without initial consent. Therefore, these ethical concerns need to be carefully addressed to ensure responsible and fair use. The ML models must be supplemented with explainable interpretation tools that can precisely explain decisions in understandable terms. Lastly, they must be implemented with informed consent, traceable logs, robust cybersecurity and final human approvals.

One of the primary challenges in the application of ML algorithms in bioprocesses is that it demands deep interdisciplinary knowledge, which must be a blend of microbiology, biotechnology, bioprocess engineering and data science. The domain experts may lack data science skills, while data scientists may not understand the nuances of bioprocesses. The resultant skills gap can result in the generation of suboptimal models that either fail to capture biological reality or are not technically feasible for real-world use. Also, many ML models are difficult to interpret and provide predictions without clear explanations. This lack of transparency is problematic in bioprocess environments that

Table 6 Patents reported on the utilization of machine learning algorithms in bioprocess technologies

Patent Number	Publication date	Description	Reference
US12020494B2	2024-06	This US patent study reported involved using stem cells for cell therapy applications, cytotoxicity testing and drug discovery. Previously, many methods were developed for determining the characteristics of cells, but those methods resulted in the complete destruction of cells being observed, contamination issues and the generation of incomplete information on cell health. ML-based neural networks were used as a non-invasive technique for the prediction of essential attributes of different cell types. Images of the cells were taken to identify the essential characteristics, and the absorbance was used as a measure of light to generate the data for the training of a deep neural network.	Bharti et al. (2024)
JP7092879B2	2022-06	ML-based prediction of the different metabolic states of the microbial culture. The study involved the creation of metabolic fingerprints, which focused on the selection of the desired clones for increasing the product yields during fermentation. A combined approach using a neural network ML algorithm and metabolic flux analysis was reported for determining the metabolic status of cell culture.	Wolfgang et al. (2022)
US 20,220,228,102 A1	2022-07	Utilization of different machine learning algorithms for selecting the most suitable cell line for recombinant therapeutic protein production. The ML methods utilized were ridge regression, Lasso regression, PCA, PLS, Random Forest and XGBoost algorithms. The patent study reported the criteria for the selection of the most appropriate cell line for drug manufacturing using ML techniques.	Le et al. (2022)
US20210149361 A1	2021-05	A method for real-time monitoring and determination of concentration, potency and purity of a product was reported. Multivariate data analysis using several different ML techniques, including PLS, PCA, RF, NN, Multivariate adaptive regression, and SVM, was tested on the experimental data obtained from online and in-line sensors. Multivariate adaptive regression was found to be optimum for predicting the output in terms of accurately determining potency, concentration and purity in an antibody production process.	Jungbauer et al. (2021)
US20230407409A1	2023-12	Different ML classification techniques, including SVM and kNN, were applied to the experimental data on microbiomes with the aim of enhancing the yield of the product in the fermentation processes.	Becares and Fernandez (2023)
CN113240114B	2023-12	ANN model was created to decipher the relationships between the key critical parameters in a biofuel production process.	Xu (2023)
CN105259827A	2018-08	ANN in combination with fuzzy logic was utilized for the construction of a soft sensor-based expert decision system for monitoring the solid-state fermentation process.	Xiaobo et al. (2018)

require clear scientific understanding, troubleshooting capabilities, and compliance with regulatory standards.

Explainable AI (XAI) methods for transparent decision-making

Unlike most of the ML algorithms that work as black box models and decision making remains hidden, XAI methods provide insights into how the predictions are carried out. The merits include increased transparency, accountability and reliability. The XAI process ensures the effectiveness of the AI model through informed decisions. Every ML-driven decision is logged, making the process simpler to understand and justify. One such XAI technique is SHAP (Shapley additive explanations). SHAP (Shapley Additive Explanations) is an advanced machine learning technique that provides clear and mathematically sound interpretations of predictions made by an AI model. SHAP assigns Shapley values, which can quantify the importance of each input variable influencing a particular model output (Agrawal et al. 2024). SHAP not only ranks the most important variables but can also explain whether the influence of variables is positive or negative for a specific decision. This is critically important in bioprocess technologies where the outcome of the process is dependent on different process parameters. In a recent study on caproic acid and lactic acid co-production, SHAP analysis revealed the enhanced production of caproic acid upon the addition of glucose (Ma et al. 2024). In another study, SHAP analysis was utilized to accurately predict the yield of the biodiesel production process (Agrawal et al. 2024). LIME (Local Interpretable Model-Agnostic Explanations) is another advanced XAI technique that explains why an AI model made a specific prediction for an individual data point rather than considering the model as a whole. In contrast to SHAP, LIME explains individual (instance) predictions from the utilized ML models (Walther et al. 2023).

Future prospects of machine learning in bioprocessing

Machine learning has revolutionized the field of bioprocessing through efficient process optimization, automation, process control, quality control and sustainable resource

utilization. The ultimate outcomes include the enhanced yield and productivity of the final product, enhanced product quality and operational consistency. Additionally, the predictive quality analytics of ML can forecast the quality of the final product based on historical data and thus allow for proactive measures to address any quality issues in advance. ML can quickly analyze large datasets to identify patterns that might be missed by human analysts, resulting in more precise and optimized bioprocesses.

The future prospects of machine learning in bioprocessing lie in the development of more robust soft sensors which can enable continuous monitoring of a greater number of critical process parameters under real-time conditions. By predicting future process states, soft sensors enable advanced control strategies like Model Predictive Control (MPC), which can anticipate and mitigate potential issues before they impact the process. Also, the combination of machine learning and IoT devices will create more advanced monitoring systems that can provide real-time data and analytics and improve the overall process control. Wireless sensor networks enabled by machine learning offer continuous monitoring and real-time adjustments, which will further enhance the efficiency and reliability of bioprocesses. In the future, ML can automate compliance checks and documentation in order to ensure that the bioprocesses meet regulatory standards more efficiently. Cloud-based ML platforms will facilitate collaboration among researchers, industry professionals, and institutions, leading to accelerated innovation. The emerging technologies incorporating machine learning in bioprocessing include the ML-powered soft sensors, multi-omics data integration and Explainable Artificial Intelligence (XAI) models and systems, which can lead to a more reliable, faster and cost-effective biomanufacturing.

Conclusion

The diverse ML algorithms possess the immense capabilities to tackle the non-linear relationships and meaningful patterns that are incomprehensively interpreted using conventional methods. ML algorithms can be readily integrated in diverse bioprocess technologies for enhancing product yield and productivity through optimization, precisely forecasting the system behaviour, predicting the uncertainties and preventing unnecessary downstream processing operations beforehand. ML algorithms can efficiently model scale-up processes, perform continuous monitoring of critical variables in real time, detection of failures to avoid batch failures and recommend corrective solutions. With the continuous evolution in sensor technology, automation and computational infrastructure, the integration of machine learning in bioprocesses can pave the way for a data-driven approach that can provide better adaptability to bioprocess dynamics and lead to sustainable biomanufacturing. However, the powerful capabilities offered by machine learning in bioprocess industries also introduce computational challenges that cannot be overlooked. These include the need for high-performance hardware, vast data storage, and robust integration frameworks. Ultimately, the successful deployment of machine learning in a typical industrial setting will depend on a balanced approach that combines scalable infrastructure, efficient algorithms, and practical workflows.

Author contributions All authors contributed to the study conception and design. D.S. had the idea for the article. Literature search, Data collection and analysis were performed by D.S. and K.S. The first draft of the manuscript was written by D.S. and K.S. critically revised the manuscript. All authors read and approved the final manuscript.

Funding No funding was received for conducting this study.

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

References

Abbasi M, Pishvaee MS, Mohseni S (2021) Third-generation biofuel supply chain: A comprehensive review and future research directions. J Clean Prod 323:129100. https://doi.org/10.1016/j.clepro.2021.129100

Acosta-Pavas JC, Robles-Rodriguez CE, Griol D, Daboussi F, Aceves-Lara CA, Corrales DC (2024) Soft sensors based on interpretable learners for industrial-scale fed-batch fermentation: learning from simulations. Comput Chem Eng 187:108736. https://doi.org/10.1016/j.compchemeng.2024.108736

Adeleke AA, Okolie JA, Ogbaga CC, Ikubanni PP, Okoye PU, Akande O (2024) Machine learning model for the evaluation of biomethane potential based on the biochemical composition of biomass. Bioenergy Res 17:731–743. https://doi.org/10.1007/s12155-023-10681-9

Agarwal H, Rathore AS, Hadpe SR, Alva SJ (2016) Artificial neural network (ANN)-based prediction of depth filter loading capacity for filter sizing. Biotechnol Prog 32:1436–1443. https://doi.org/10.1002/btpr.2329

Aghbashlo M, Peng W, Tabatabaei M, Kalogirou SA, Soltanian S, Hosseinzadeh-Bandbafha H, Mahian O, Lam SS (2021) Machine learning technology in biodiesel research: A review. Prog Energy Combust Sci 85:100904. https://doi.org/10.1016/j.pecs.2021.100904

Agrawal P, Gnanaprakash R, Dhawane SH (2024) Prediction of biodiesel yield employing machine learning: interpretability analysis via Shapley additive explanations. Fuel 359:130516. https://doi.org/10.1016/j.fuel.2023.130516

Alam MN, Anupa A, Kodamana H, Rathore AS (2024) A deep learning-aided multi-objective optimization of a downstream process for production of monoclonal antibody products. Biochem Eng J 208:109357. https://doi.org/10.1016/j.bej.2024.109357

Althuri A, Gujjala LKS, Banerjee R (2017) Partially consolidated bioprocessing of mixed lignocellulosic feedstocks for ethanol production. Bioresour Technol 245:530–539. https://doi.org/10.1016/j.biortech.2017.08.140

Alzubaidi L, Zhang J, Humaidi AJ, Al-Dujaili A, Duan Y, Al-Shamma O, Santamaría J, Fadhel MA, Al-Amidie M, Farhan L (2021) Review of deep learning: concepts, CNN architectures, challenges, applications, future directions. J Big Data 8:1–74. https://doi.org/10.1186/s40537-021-00444-8

Ambaye TG, Vaccari M, Bonilla-Petriciolet A, Prasad S, van Hullebusch ED, Rtimi S (2021) Emerging technologies for biofuel production: A critical review on recent progress, challenges and perspectives. J Environ Manage 290:112627. https://doi.org/10.1016/j.jenvman.2021.112627

Amenaghawon AN, Odika P, Aiwekhoe SE (2022) Optimization of nutrient medium composition for the production of lipase from waste cooking oil using response surface methodology and artificial neural networks. Chem Eng Commun 209:1531–1541. https://doi.org/10.1080/00986445.2021.1980395

Amenaghawon AN, Igemhokhai S, Eshiemogie SA, Ugbodu F, Evbarunegbe NI (2024) Data-driven intelligent modeling, optimization, and global sensitivity analysis of a Xanthan gum biosynthesis process. https://doi.org/10.1016/j.heliyon.2024.e25432. Heliyon 10

Asitok A, Ekpenyong M, Ben U, Antigha R, Ogarekpe N, Rao A, Akpan A, Benson N, Essien J, Antai S (2023) Stochastic modeling and meta-heuristic multivariate optimization of bioprocess conditions for co-valorization of feather and waste frying oil toward prodigiosin production. Prep Biochem Biotechnol 53:690–703. https://doi.org/10.1080/10826068.2022.2134891

Asrul MAM, Atan MF, Yun HAH, Lai JCH (2022) A review of advanced optimization strategies for fermentative biohydrogen production processes. Int J Hydrogen Energy 47:16785–16804. https://doi.org/10.1016/j.ijhydene.2022.03.197

Bandpey AF, Abdi J, Firozjaee TT (2024) Improved Estimation of dark fermentation biohydrogen production utilizing a robust categorical boosting machine-learning algorithm. Int J Hydrogen Energy 52:190–199. https://doi.org/10.1016/j.ijhydene.2023.11.137

Bapat PM, Wangikar PP (2004) Optimization of rifamycin B fermentation in shake flasks via a machine-learning-based approach. Biotechnol Bioeng 86:201–208. https://doi.org/10.1002/bit.20056

Bardhan SK, Gupta S, Gorman ME, Haider MA (2015) Biorenewable chemicals: feedstocks, technologies and the conflict with food production. Renew Sustain Energy Rev 51:506–520. https://doi.org/10.1016/j.rser.2015.06.013

Becares AA, Fernandez AF (2023) Microbiome based identification, monitoring and enhancement of fermentation processes and products https://patents.google.com/patent/US20230407409A1/en?oq=US20230407409A1 Accessed 2 Jan 2025

Bedi S, Samal A, Ray C, Snow D (2020) Comparative evaluation of machine learning models for groundwater quality assessment. Environ Monit Assess 192:1–23. https://doi.org/10.1007/s10661-020-08695-3

Beig B, Riaz M, Naqvi SR, Hassan M, Zheng Z, Karimi K, Pugazhendhi A, Atabani AE, Chi NTL (2021) Current challenges and innovative developments in pretreatment of lignocellulosic residues for biofuel production: A review. Fuel 287:119670. https://doi.org/10.1016/j.fuel.2020.119670

Bezerra CO, Carneiro LL, Carvalho EA, das Chagas TP, de Carvalho LR, Uetanabaro APT, da Silva GP, da Silva EGP, da Costa AM (2021) Artificial intelligence as a combinatorial optimization strategy for cellulase production by *Trichoderma stromaticum* AM7 using peach-palm waste under solid-state fermentation. BioEnergy Res 14:1161–1170. https://doi.org/10.1007/s12155-020-10234-4

Bhargava A, Bansal A, Goyal V, Shukla A (2023) Machine learning & computer vision-based optimum black tea fermentation detection. Multimed Tools Appl 82:43335–43347. https://doi.org/10.1007/s11042-023-15453-3

Bharti K, Hotaling NA, SCHAUB NJ (2024) SIMON CG using machine learning and/or neural networks to validate stem cells and their derivatives (2-D cells and 3-D tissues) for use in cell therapy and tissue engineered products. https://patents.google.com/patent/US12020494B2/en?oq=US12020494B2 Accessed 2 Jan 2025

Bhatia L, Bachheti RK, Garlapati VK, Chandel AK (2022) Third-generation biorefineries: a sustainable platform for food, clean energy, and nutraceuticals production. Biomass Conv Bioref 12:4215–4230. https://doi.org/10.1007/s13399-020-00843-6

Bonanni D, Litrico M, Ahmed W, Morerio P, Cazzorla T, Spaccapaniccia E, Cattani F, Allegretti M, Beccari AR, Del Bue A, Martin F (2023) A deep learning approach to optimize Recombinant protein production in *Escherichia coli* fermentations. Fermentation 9:503. https://doi.org/10.3390/fermentation9060503

Boodhoo KVK, Flickinger MC, Woodley JM, Emanuelsson EAC (2022) Bioprocess intensification: A route to efficient and sustainable biocatalytic transformations for the future. Chem Eng Process -Process Intensif 172:108793. https://doi.org/10.1016/j.jcep.2022.108793

Bowler AL, Pound MP, Watson NJ (2021) Domain adaptation and federated learning for ultrasonic monitoring of beer fermentation. Fermentation 7:253. https://doi.org/10.3390/fermentation7040253

Bradford E, Schweidtmann AM, Zhang D, Jing K, del Rio-Chanona EA (2018) Dynamic modeling and optimization of sustainable algal production with uncertainty using multivariate Gaussian processes. Comput Chem Eng 118:143–158. https://doi.org/10.1016/j.compchemeng.2018.07.015

Buller R, Lutz S, Kazlauskas RJ, Snajdrova R, Moore JC, Bornscheuer UT (2023) From nature to industry: Harnessing enzymes for biocatalysis. Science 382:8615. https://doi.org/10.1126/science.adh8615

Cadet F, Fontaine N, Li G, Sanchis J, Ng Fuk Chong M, Pandjaitan R, Vetrivel I, Offmann B, Reetz MT (2018) A machine learning approach for reliable prediction of amino acid interactions and its application in the directed evolution of enantioselective enzymes. Sci Rep 8:16757. https://doi.org/10.1038/s41598-018-35033-y

Chen C, Wong HE, Goudar CT (2018) Upstream process intensification and continuous manufacturing. Curr Opin Chem Eng 22:191–198. https://doi.org/10.1016/j.coche.2018.10.006

Ching PM, So RH, Morck T (2021) Advances in soft sensors for wastewater treatment plants: A systematic review. J Water Process Eng 44:102367. https://doi.org/10.1016/j.jwpe.2021.102367

Ching PML, Zou X, Wu D, So RHY, Chen GH (2022) Development of a wide-range soft sensor for predicting wastewater BOD5 using an eXtreme gradient boosting (XGBoost) machine. Environ Res 210:112953. https://doi.org/10.1016/j.envres.2022.112953

Coşgun A, Günay ME, Yıldırım R (2021) Exploring the critical factors of algal biomass and lipid production for renewable fuel production by machine learning. Renew Energy 163:1299–1317. https://doi.org/10.1016/j.renene.2020.09.034

Coşgun A, Günay ME, Yıldırım R (2022) Analysis of lipid production from Yarrowia lipolytica for renewable fuel production by machine learning. Fuel 315:122817. https://doi.org/10.1016/j.fuel.2021.122817

Cruz IA, Chuenchart W, Long F, Surendra KC, Andrade LRS, Bilal M, Liu H, Figueiredo RT, Khanal SK, Ferreira LFR (2022) Application of machine learning in anaerobic digestion: perspectives and challenges. Bioresour Technol 345:126433. https://doi.org/10.1016/j.biortech.2021.126433

Dave N, Varadavenkatesan T, Selvaraj R, Vinayagam R (2021) Modelling of fermentative bioethanol production from Indigenous Ulva prolifera biomass by *Saccharomyces cerevisiae* NFCCI1248 using an integrated ANN-GA approach. Sci Total Environ 791:148429. https://doi.org/10.1016/j.scitotenv.2021.148429

David AJ, Abinandan S, Vaidyanathan VK, Xu CC, Krishnamurthi T (2023) A critical review on current status and environmental sustainability of pre-treatment methods for bioethanol production from lignocellulose feedstocks. 3 Biotech 13:233. https://doi.org/10.1007/s13205-023-03657-1

De Clercq D, Wen Z, Fei F, Caicedo L, Yuan K, Shang R (2020) Interpretable machine learning for predicting biomethane production in industrial-scale anaerobic co-digestion. Sci Total Environ 712:134574. https://doi.org/10.1016/j.scitotenv.2019.134574

De Diego IM, Redondo AR, Fernández RR, Navarro J, Moguerza JM (2022) General performance score for classification problems. Appl Intell 52:12049–12063. https://doi.org/10.1007/s10489-021-03041-7

de Mello AFM, de Souza Vandenberghe LP, Herrmann LW, Letti LAJ, Burgos WJM, Scapini T, Manzoki MC, de Oliveira PZ, Soccol CR (2024) Strategies and engineering aspects on the scale-up of bioreactors for different bioprocesses. Syst Microbiol Biomanufacturing 4:365–385. https://doi.org/10.1007/s43393-023-00205-z

de Menezes LHS, Carneiro LL, de Carvalho Tavares IM, Santos PH, das Chagas TP, Mendes AA, da Silva EGP, Franco M, de Oliveira JR (2021) Artificial neural network hybridized with a genetic algorithm for optimization of lipase production from *Penicillium roqueforti* ATCC 10110 in solid-state fermentation. Biocatal Agric Biotechnol 31:101885. https://doi.org/10.1016/j.bcab.2020.101885

de Oliveira RA, Komesu A, Rossell CEV, Maciel Filho R (2018) Challenges and opportunities in lactic acid bioprocess design—From economic to production aspects. Biochem Eng J 133:219–239. https://doi.org/10.1016/j.bej.2018.03.003

del Rio-Chanona EA, Manirafasha E, Zhang D, Yue Q, Jing K (2016) Dynamic modeling and optimization of cyanobacterial C-phycocyanin production process by artificial neural network. Algal Res 13:7–15. https://doi.org/10.1016/j.algal.2015.11.004

del Rio-Chanona EA, Wagner JL, Ali H, Fiorelli F, Zhang D, Hellgardt K (2019) Deep learning-based surrogate modeling and optimization for microalgal biofuel production and photobioreactor design. AIChE J 65:915–923. https://doi.org/10.1002/aic.16473

Ding H, Tian J, Yu W, Wilson DI, Young BR, Cui X, Xin X, Wang Z, Li W (2023) The application of artificial intelligence and big data in the food industry. Foods 12:4511. https://doi.org/10.3390/foods12244511

Dixit M, Chhabra D, Shukla P (2023) Optimization of endoglucanase-lipase-amylase enzyme consortium from *Thermomyces lanuginosus* VAPS25 using Multi-Objective genetic algorithm and their bio-deinking applications. Bioresour Technol 370:128467. https://doi.org/10.1016/j.biortech.2022.128467

Dudala SS, TC V (2024) Modeling and optimization of uricase production from a novel Pseudomonas mosselii using response surface methodology and artificial neural network. Biomass Conv Bioref 14:21865–21880. https://doi.org/10.1007/s13399-023-04468-3

Dürauer A, Jungbauer A, Scharl T (2024) Sensors and chemometrics in downstream processing. Biotechnol Bioeng 121:2347–2364. https://doi.org/10.1002/bit.28499

Durrant JD, Amaro RE (2015) Machine-learning techniques applied to antibacterial drug discovery. Chem Biol Drug Des 85:14–21. https://doi.org/10.1111/cbdd.12423

Ekpenyong M, Asitok A, Antai S, Ekpo B, Antigha R, Ogarekpe N (2021) Statistical and artificial neural network approaches to modeling and optimization of fermentation conditions for production of a surface/bioactive glyco-lipo-peptide. Int J Pept Res Ther 27:475–495. https://doi.org/10.1007/s10989-020-10094-8

Feng Y, Wang Y, Beykal B, Qiao M, Xiao Z, Luo Y (2024) A mechanistic review on machine learning-supported detection and analysis of volatile organic compounds for food quality and safety. Trends Food Sci Technol 143:104297. https://doi.org/10.1016/j.tifs.2023.104297

Galvanauskas V, Simutis R, Lübbert A (2004) Hybrid process models for process optimisation, monitoring and control. Bioprocess Biosyst Eng 26:393–400. https://doi.org/10.1007/s00449-004-0385-x

Gangadharan N, Sewell D, Turner R, Field R, Cheeks M, Oliver SG, Slater NK, Dikicioglu D (2021) Data intelligence for process performance prediction in biologics manufacturing. Comput Chem Eng 146:107226. https://doi.org/10.1016/j.compchemeng.2021.107226

Garg A, Jain S (2020) Process parameter optimization of biodiesel production from algal oil by response surface methodology and artificial neural networks. Fuel 277:118254. https://doi.org/10.1016/j.fuel.2020.118254

Gerstweiler L, Bi J, Middelberg AP (2021) Continuous downstream bioprocessing for intensified manufacture of biopharmaceuticals and antibodies. Chem Eng Sci 231:116272. https://doi.org/10.1016/j.ces.2020.116272

Gonzalez Viejo C, Torrico DD, Dunshea FR, Fuentes S (2019) Emerging technologies based on artificial intelligence to assess the quality and consumer preference of beverages. Beverages 5:62. https://doi.org/10.3390/beverages5040062

Goodswen SJ, Barratt JL, Kennedy PJ, Kaufer A, Calarco L, Ellis JT (2021) Machine learning and applications in microbiology. FEMS Microbiol Rev 45:fuab015. https://doi.org/10.1093/femsre/fuab015

Gopakumar V, Tiwari S, Rahman I (2018) A deep learning based data driven soft sensor for bioprocesses. Biochem Eng J 136:28–39. https://doi.org/10.1016/j.bej.2018.04.015

Gottumukkala LD, Haigh K, Görgens J (2017) Trends and advances in conversion of lignocellulosic biomass to biobutanol: microbes, bioprocesses and industrial viability. Renew Sustain Energy Rev 76:963–973. https://doi.org/10.1016/j.rser.2017.03.030

Helleckes LM, Hemmerich J, Wiechert W, von Lieres E, Grünberger A (2023) Machine learning in bioprocess development: from promise to practice. Trends Biotechnol 41:817–835. https://doi.org/10.1016/j.tibtech.2022.10.010

Hlangwani E, Doorsamy W, Adebiyi JA, Fajimi LI, Adebo OA (2021) A modeling method for the development of a bioprocess to optimally produce Umqombothi (a South African traditional beer). Sci Rep 11:20626. https://doi.org/10.1038/s41598-021-00097-w

Holzinger A, Keiblinger K, Holub P, Zatloukal K, Müller H (2023) AI for life: trends in artificial intelligence for biotechnology. New Biotechnol 74:16–24. https://doi.org/10.1016/j.nbt.2023.02.001

Hosseinzadeh A, Zhou JL, Altaee A, Li D (2022) Machine learning modeling and analysis of biohydrogen production from wastewater by dark fermentation process. Bioresour Technol 343:126111. https://doi.org/10.1016/j.biortech.2021.126111

Jia R, Song YC, Piao DM, Kim K, Lee CY, Park J (2022) Exploration of deep learning models for real-time monitoring of state and performance of anaerobic digestion with online sensors. Bioresour Technol 363:127908. https://doi.org/10.1016/j.biortech.2022.127908

Jin H, Kim YG, Jin Z, Rushchitc AA, Al-Shati AS (2022) Optimization and analysis of bioenergy production using machine learning modeling: Multi-layer perceptron, Gaussian processes regression, K-nearest neighbors, and artificial neural network models. Energy Rep 8:13979–13996. https://doi.org/10.1016/j.egyr.2022.10.334

Jungbauer A, Durauer A, Walch N, Sauer D, Scharl-Hirsch T, Melcher M, Leisch F (2021) Real time monitoring of product purification.

https://patents.google.com/patent/US20210149361A1/en?oq=US20210149361+A1 Accessed 2 Jan 2025

Kabli M, Yilmaz MT, Taylan O, Kaya Y, İspirli H, Basahel A, Sagdic O, Dertli E (2020) An integrated neural-fuzzy methodology for characterisation and modelling of exopolysaccharide (EPS) production levels of *Leuconostoc mesenteroides* DL1. Comput Ind Eng 148:106619. https://doi.org/10.1016/j.cie.2020.106619

Kalwani M, Kumari A, Rudra SG, Chhabra D, Pabbi S, Shukla P (2024) Application of ANN-MOGA for nutrient sequestration for wastewater remediation and production of polyunsaturated fatty acid (PUFA) by *Chlorella sorokiniana* MSP1. Chemosphere 349:140835. https://doi.org/10.1016/j.chemosphere.2023.140835

Kamal I, Ashfaq UA, Hayat S, Aslam B, Sarfraz MH, Yaseen H, Rajoka MSR, Shah AA, Khurshid M (2023) Prospects of antimicrobial peptides as an alternative to chemical preservatives for food safety. Biotechnol Lett 45:137–162. https://doi.org/10.1007/s10529-022-03328-w

Kang CK, Shin J, Cha Y, Kim MS, Choi MS, Kim T, Park YK, Choi YJ (2023) Machine learning-guided prediction of potential engineering targets for microbial production of lycopene. Bioresour Technol 369:128455. https://doi.org/10.1016/j.biortech.2022.128455

Karimi Alavijeh M, Lee YY, Gras SL (2024) A perspective-driven and technical evaluation of machine learning in bioreactor scale-up: A case-study for potential model developments. Eng Life Sci 24:2400023. https://doi.org/10.1002/elsc.202400023

Katongtung T, Onsree T, Tippayawong N (2022) Machine learning prediction of biocrude yields and higher heating values from hydrothermal liquefaction of wet biomass and wastes. Bioresour Technol 344:126278. https://doi.org/10.1016/j.biortech.2021.126278

Kavitha S, Ravi YK, Kumar G, Nandabalan YK (2024) Microalgal biorefineries: advancement in machine learning tools for sustainable biofuel production and value-added products recovery. J Environ Manage 353:120135. https://doi.org/10.1016/j.jenvman.2024.120135

Khuat TT, Bassett R, Otte E, Grevis-James A, Gabrys B (2024) Applications of machine learning in antibody discovery, process development, manufacturing and formulation: current trends, challenges, and opportunities. Comput Chem Eng 182:108585. https://doi.org/10.1016/j.compchemeng.2024.108585

Khurshid A, Pani AK (2023) Machine learning approaches for data-driven process monitoring of biological wastewater treatment plant: A review of research works on benchmark simulation model. Environ Monit Assess 195 1:916. https://doi.org/10.1007/s10661-023-11463-8

Kirdar AO, Chen G, Weidner J, Rathore AS (2010) Combining near-infrared (NIR) spectroscopy and multivariate data analysis (MVDA) for screening of Raw materials used in the cell culture medium for the production of a Recombinant therapeutic protein. Biotechnol Prog 26:527–531. https://doi.org/10.1002/btpr.329

Konishi M (2020) Bioethanol production estimated from volatile compositions in hydrolysates of lignocellulosic biomass by deep learning. J Biosci Bioeng 129:723–729. https://doi.org/10.1016/j.jbiosc.2020.01.006

Kotidis P, Kontoravdi C (2020) Harnessing the potential of artificial neural networks for predicting protein glycosylation. Metab Eng Commun 10:00131. https://doi.org/10.1016/j.mec.2020.e00131

Krippl M, Dürauer A, Duerkop M (2020a) Hybrid modeling of cross-flow filtration: predicting the flux evolution and duration of ultrafiltration processes. Sep Purif Technol 248:117064. https://doi.org/10.1016/j.seppur.2020.117064

Krippl M, Bofarull-Manzano I, Duerkop M, Dürauer A (2020b) Hybrid modeling for simultaneous prediction of flux, rejection factor and concentration in two-component crossflow ultrafiltration. Processes 8:1625. https://doi.org/10.3390/pr8121625

Kumar B, Verma P (2021) Biomass-based biorefineries: an important architype towards a circular economy. Fuel 288:119622. https://doi.org/10.1016/j.fuel.2020.119622

Kumar M, Sun Y, Rathour R, Pandey A, Thakur IS, Tsang DC (2020) Algae as potential feedstock for the production of biofuels and value-added products: opportunities and challenges. Sci Total Environ 716:137116. https://doi.org/10.1016/j.scitoenv.2020.137116

Kyaw KS, Adegoke SC, Ajani CK, Nwabor OF, Onyeaka H (2024) Toward in-process technology-aided automation for enhanced microbial food safety and quality assurance in milk and beverages processing. Crit Rev Food Sci Nutr 64:1715–1735. https://doi.org/10.1080/10408398.2022.2118660

Lahiri D, Nag M, Dutta B, Sarkar T, Ray RR (2021) Artificial neural network and response surface methodology-mediated optimization of bacteriocin production by *Rhizobium leguminosarum*. Iran J Sci Technol Trans A: Sci 45:1509–1517. https://doi.org/10.1007/s40995-021-01157-6

Lau HL, Wong FWF, Abd Rahman RNZR, Mohamed MS, Ariff AB, Hii SL (2023) Optimization of fermentation medium components by response surface methodology (RSM) and artificial neural network hybrid with genetic algorithm (ANN-GA) for lipase production by *Burkholderia cenocepacia* ST8 using used automotive engine oil as substrate. Biocatal Agric Biotechnol 50:102696. https://doi.org/10.1016/j.bcab.2023.102696

Laurence A, Sahu N, Mahanty B (2024) Artificial neural network based optimization of sunflower oil supplementation in polyhydroxyalkanoates production by *Cupriavidus necator*. Biocatal Agric Biotechnol 59:103281. https://doi.org/10.1016/j.bcab.2024.103281

Lawson CE, Martí JM, Radivojevic T, Jonnalagadda SVR, Gentz R, Hillson NJ, Peisert S, Kim J, Simmons BA, Petzold CJ, Singer SW (2021) Machine learning for metabolic engineering: A review. Metab Eng 63:34–60. https://doi.org/10.1016/j.ymben.2020.10.005

Le KH, Xie Y, STEVENS JL, Baskerville-Bridges A (2022) Data-driven predictive modeling for cell line selection in biopharmaceutical production. https://patents.google.com/patent/US20220228102A1/en?oq=US+20220228102+A1 Accessed 2 Jan 2025

Lhamo P, Mahanty B, Behera SK (2023) Optimization of biomass and polyhydroxyalkanoate production by *Cupriavidus necator* using response surface methodology and genetic algorithm optimized artificial neural network. Biomass Conv Bioref 14:20053–20068. https://doi.org/10.1007/s13399-023-04043-w

Li M, Wijewardane NK, Ge Y, Xu Z, Wilkins MR (2020) Visible/near infrared spectroscopy and machine learning for predicting polyhydroxybutyrate production cultured on alkaline pretreated liquor from corn Stover. Bioresour Technol Rep 9:100386. https://doi.org/10.1016/j.biteb.2020.100386

Li J, Zhang W, Liu T, Yang L, Li H, Peng H, Jiang S, Wang X, Leng L (2021) Machine learning aided bio-oil production with high energy recovery and low nitrogen content from hydrothermal liquefaction of biomass with experiment verification. Chem Eng J 425:130649. https://doi.org/10.1016/j.cej.2021.130649

Li H, Qiu T, You F (2024) AI-based optimal control of fed-batch biopharmaceutical process leveraging deep reinforcement learning. Chem Eng Sci 292:119990. https://doi.org/10.1016/j.ces.2024.119990

Liu G, Stokes JM (2022) A brief guide to machine learning for antibiotic discovery. Curr Opin Microbiol 69:102190. https://doi.org/10.1016/j.mib.2022.102190

Liyanaarachchi VC, Nishshanka GKSH, Sakarika M, Nimarshana PHV, Ariyadasa TU, Kornaros M (2021) Artificial neural network (ANN) approach to optimize cultivation conditions of microalga *Chlorella vulgaris* in view of biodiesel production. Biochem Eng J 173:108072. https://doi.org/10.1016/j.bej.2021.108072

Lu NV, Tansuchat R, Yuizono T, Huynh VN (2020) Incorporating active learning into machine learning techniques for sensory evaluation of food. Int J Comput Intell Syst 13:655–662. https://doi.org/10.2991/ijcis.d.200525.001

Lu K, Liu L, Xu Z, Xie W (2022) The analysis of volatile compounds through flavoromics and machine learning to identify the origin of traditional Chinese fermented shrimp paste from different regions. Lwt 171:114096. https://doi.org/10.1016/j.lwt.2022.114096

Ma Y, Noreña-Caro DA, Adams AJ, Brentzel TB, Romagnoli JA, Benton MG (2020) Machine-learning-based simulation and fed-batch control of cyanobacterial-phycocyanin production in *Plectonema* by artificial neural network and deep reinforcement learning. Comput. Chem. Eng. 142:107016. https://doi.org/10.1016/j.compchemeng.2020.107016

Ma EJ, Siirola E, Moore C, Kummer A, Stoeckli M, Faller M, Bouquet C, Eggimann F, Ligibel M, Huynh D, Cutler G (2021) Machine-directed evolution of an Imine reductase for activity and stereoselectivity. ACS Catal 11:12433–12445. https://doi.org/10.1021/acscatal.1c02786

Ma H, Liu Y, Zhao J, Fei F, Gao M, Wang Q (2024) Explainable machine learning-driven predictive performance and process parameter optimization for Caproic acid production. Bioresour Technol 410:131311. https://doi.org/10.1016/j.biortech.2024.131311

Mahapatra S, Kumar D, Singh B, Sachan PK (2021) Biofuels and their sources of production: A review on cleaner sustainable alternative against conventional fuel, in the framework of the food and energy nexus. Energy Nexus 4:100036. https://doi.org/10.1016/j.nexus.2021.100036

Mahmoodi-Eshkaftaki M, Mockaitis G, Rafiee MR (2024) Dynamic optimization of volatile fatty acids to enrich biohydrogen production using a deep learning neural network. Biomass Conv Bioref 14:8003–8014. https://doi.org/10.1007/s13399-022-02899-y

Manikandan S, Subbaiya R, Biruntha M, Krishnan RY, Muthusamy G, Karmegam N (2022) Recent development patterns, utilization and prospective of biofuel production: emerging Nanotechnological intervention for environmental sustainability–A review. Fuel 314:122757. https://doi.org/10.1016/j.fuel.2021.122757

Mikulčić H, Baleta J, Zhang Z, Klemeš JJ (2023) Sustainable development of energy, water and environmental systems in the changing world. J Clean Prod 390:135945. https://doi.org/10.1016/j.jclepro.2023.135945

Mirakhori F, Niazi SK (2025) Harnessing the AI/ML in drug and biological products discovery and development: the regulatory perspective. Pharmaceuticals 18:47. https://doi.org/10.3390/ph18010047

Mondal PP, Galodha A, Verma VK, Singh V, Show PL, Awasthi MK, Lall B, Anees S, Pollmann K, Jain R (2023) Review on machine learning-based bioprocess optimization, monitoring, and control systems. Bioresour Technol 370:128523. https://doi.org/10.1016/j.biortech.2022.128523

Monroy I, Guevara-López E, Buitrón G (2018) Biohydrogen production by batch indoor and outdoor photo-fermentation with an immobilized consortium: a process model with neural networks. Biochem Eng J 135:1–10. https://doi.org/10.1016/j.bej.2018.01.026

Mowbray M, Savage T, Wu C, Song Z, Cho BA, Del Rio-Chanona EA, Zhang D (2021) Machine learning for biochemical engineering: A review. Biochem Eng J 172:108054. https://doi.org/10.1016/j.bej.2021.108054

Munappy AR, Bosch J, Olsson HH, Arpteg A, Brinne B (2022) Data management for production quality deep learning models: challenges and solutions. J Syst Softw 191:111359. https://doi.org/10.1016/j.jss.2022.111359

Nandi S, Subramanian A, Sarkar RR (2017) An integrative machine learning strategy for improved prediction of essential genes in *Escherichia coli* metabolism using flux-coupled features. Mol Biosyst 13:1584–1596. https://doi.org/10.1039/C7MB00234C

Narayanan H, Sokolov M, Morbidelli M, Butté A (2019) A new generation of predictive models: the added value of hybrid models for manufacturing processes of therapeutic proteins. Biotechnol Bioeng 116:2540–2549. https://doi.org/10.1002/bit.27097

Narayanan H, Luna MF, von Stosch M, Cruz Bournazou MN, Polotti G, Morbidelli M, Butté A, Sokolov M (2020) Bioprocessing in the digital age: the role of process models. Biotechnol J 15:1900172. https://doi.org/10.1002/biot.201900172

Narayanan H, Dingfelder F, Butté A, Lorenzen N, Sokolov M, Arosio P (2021) Machine learning for biologics: opportunities for protein engineering, developability, and formulation. Trends Pharmacol Sci 42:151–165. https://doi.org/10.1016/j.tips.2020.12.004

Naser MZ, Alavi AH (2023) Error metrics and performance fitness indicators for artificial intelligence and machine learning in engineering and sciences. Arch Struct Constr 3:499–517. https://doi.org/10.1007/s44150-021-00015-8

Nguyen XDJ, Liu YA, McDowell CC, Dooley L (2025) Methodology for contamination detection and reduction in fermentation processes using machine learning. Bioprocess Biosyst Eng 1–17. https://doi.org/10.1007/s00449-025-03194-6

Niaze AA, Sahu R, Sunkara MK, Upadhyayula S (2023) Model construction and optimization for Raising the concentration of industrial bioethanol production by using a data-driven ANN model. Renew Energy 216:119031. https://doi.org/10.1016/j.renene.2023.119031

Nikita S, Thakur G, Jesubalan NG, Kulkarni A, Yezhuvath VB, Rathore AS (2022) AI-ML applications in bioprocessing: ML as an enabler of real time quality prediction in continuous manufacturing of mAbs. Comput Chem Eng 164:107896. https://doi.org/10.1016/j.compchemeng.2022.107896

Nikkhah A, Rohani A, Zarei M, Kulkarni A, Batarseh FA, Blackstone NT, Ovissipour R (2023) Toward sustainable culture media: using artificial intelligence to optimize reduced-serum formulations for cultivated meat. Sci Total Environ 894:164988. https://doi.org/10.1016/j.scitotenv.2023.164988

Okolie JA (2024) Introduction of machine learning and artificial intelligence in biofuel technology. Curr Opin Green Sust 47:100928. https://doi.org/10.1016/j.cogsc.2024.100928

Pacheco VL, Bragagnolo L, Dalla Rosa F, Thomé A (2023) Optimization of biocementation responses by artificial neural network and random forest in comparison to response surface methodology. Environ Sci Pollut Res 30:61863–61887. https://doi.org/10.1007/s11356-023-26362-1

Panjwani S, Cui I, Spetsieris K, Mleczko M, Wang W, Zou JX, Anwaruzzaman M, Liu S, Canales R, Hesse O (2021) Application of machine learning methods to pathogen safety evaluation in biological manufacturing processes. Biotechnol Prog 37:e3135. https://doi.org/10.1002/btpr.3135

Patel V, Shah M (2022) Artificial intelligence and machine learning in drug discovery and development. Intell Med 2:134–140. https://doi.org/10.1016/j.imed.2021.10.001

Patra P, Disha BR, Kundu P, Das M, Ghosh A (2023) Recent advances in machine learning applications in metabolic engineering. Biotechnol Adv 62:108069. https://doi.org/10.1016/j.biotechadv.2022.108069

Pereira RD, Badino AC, Cruz AJ (2020) Framework based on artificial intelligence to increase industrial bioethanol production. Energy Fuels 34:4670–4677. https://doi.org/10.1021/acs.energyfuels.0c00033

Perera YS, Ratnaweera DAAC, Dasanayaka CH, Abeykoon C (2023) The role of artificial intelligence-driven soft sensors in advanced sustainable process industries: A critical review. Eng Appl Artif

Intell 121:105988. https://doi.org/10.1016/j.engappai.2023.105988

Phromphithak S, Onsree T, Tippayawong N (2021) Machine learning prediction of cellulose-rich materials from biomass pretreatment with ionic liquid solvents. Bioresour Technol 323:124642. https://doi.org/10.1016/j.biortech.2020.124642

Puranik A, Dandekar P, Jain R (2022) Exploring the potential of machine learning for more efficient development and production of biopharmaceuticals. Biotechnol Prog 38:e3291. https://doi.org/10.1002/btpr.3291

Rabiya R, Sen R (2022) Artificial intelligence driven advanced optimization strategy vis-à-vis response surface optimization of production medium: bacterial exopolysaccharide production as a case-study. Biochem Eng J 178:108271. https://doi.org/10.1016/j.bej.2021.108271

Rathore AS, Mittal S, Lute S, Brorson K (2012) Chemometrics applications in biotechnology processes: predicting column integrity and impurity clearance during reuse of chromatography resin. Biotechnol Prog 28:1308–1314. https://doi.org/10.1002/btpr.1610

Rathore AS, Nikita S, Thakur G, Deore N (2021) Challenges in process control for continuous processing for production of monoclonal antibody products. Curr Opin Chem Eng 31:100671. https://doi.org/10.1016/j.coche.2021.100671

Rathore AS, Nikita S, Thakur G, Mishra S (2023) Artificial intelligence and machine learning applications in biopharmaceutical manufacturing. Trends Biotechnol 4:497–510. https://doi.org/10.1016/j.tibtech.2022.08.007

Rolff J, Bonhoeffer S, Kloft C, Leistner R, Regoes R, Hochberg ME (2024) Forecasting antimicrobial resistance evolution. Trends Microbiol 32:736–745. https://doi.org/10.1016/j.tim.2023.12.009

Rodgers S, Bowler A, Wells L, Lee CS, Hayes M,Poulston S, Lester E, Meng F, McKechnie J, Conradie A (2024) A surrogate model for the economic evaluation of renewable hydrogen production from feedstocks via supercritical water gasification. Int. J. Hydrogen Energy 49:277-294. https://doi.org/10.1016/j.ijhydene.2023.08.016

Saharan V, Tushir S, Singh J, Kumar N, Chhabra D, Kapoor RK (2023) Application of MOGA-ANN tool for the production of cellulase and Xylanase using de-oiled rice Bran (DORB) for bioethanol production. Biomass Conv Bioref 14:11987–11999. https://doi.org/10.1007/s13399-023-04022-1

Saini DK, Yadav D, Pabbi S, Chhabra D, Shukla P (2020) Phycobiliproteins from *Anabaena variabilis* CCC421 and its production enhancement strategies using combinatory evolutionary algorithm approach. Bioresour Technol 309:123347. https://doi.org/10.1016/j.biortech.2020.123347

Saito Y, Oikawa M, Sato T, Nakazawa H, Ito T, Kameda T, Tsuda K, Umetsu M (2021) Machine-learning-guided library design cycle for directed evolution of enzymes: the effects of training data composition on sequence space exploration. ACS Catal 11:14615–14624. https://doi.org/10.1021/acscatal.1c03753

Samuelsson J, Eiriksson FF, Åsberg D, Thorsteinsdóttir M, Fornstedt T (2019) Determining gradient conditions for peptide purification in RPLC with machine-learning-based retention time predictions. J Chromatogr A 1598:92–100. https://doi.org/10.1016/j.chroma.2019.03.043

Sarker IH (2021) Machine learning: algorithms, real-world applications and research directions. SN Comput Sci 2:160. https://doi.org/10.1007/s42979-021-00592-x

Sarker IH (2022) AI-based modeling: techniques, applications and research issues towards automation, intelligent and smart systems. SN Comput Sci 3:158. https://doi.org/10.1007/s42979-022-01043-x

Saxena A, Prasad M, Gupta A, Bharill N, Patel OP, Tiwari A, Er MJ, Ding W, Lin CT (2017) A review of clustering techniques and developments. Neurocomputing 267:664–681. https://doi.org/10.1016/j.neucom.2017.06.053

Serafim MSM, Kronenberger T, Oliveira PR, Poso A, Honorio KM, Mota BEF, Maltarollo VG (2020) The application of machine learning techniques to innovative antibacterial discovery and development. Expert Opin Drug Discov 15:1165–1180. https://doi.org/10.1016/j.mib.2022.102190

Shahrajabian MH, Sun W (2024) Assessment of wine quality, traceability and detection of grapes wine, detection of harmful substances in alcohol and liquor composition analysis. Lett Drug Des Discov 21:1377–1399. https://doi.org/10.2174/1570180820666230228115450

Sharma D, Mishra A (2022) L-asparaginase production in solid-state fermentation using *Aspergillus niger*: process modeling by artificial neural network approach. Prep Biochem Biotechnol 52:549–560. https://doi.org/10.1080/10826068.2021.1972426

Sharma D, Mishra A (2024) Synergistic effects of ternary mixture formulation and process parameters optimization in a sequential approach for enhanced L-asparaginase production using agro-industrial wastes. Environ Sci Pollut Res 31:17858–17873. https://doi.org/10.1007/s11356-023-26977-4

Sharmila VG, Shanmugavel SP, Banu JR (2024) A review on emerging technologies and machine learning approaches for sustainable production of biofuel from biomass waste. Biomass Bioenergy 180:106997. https://doi.org/10.1016/j.biombioe.2023.106997

Singh P, Pandey S, Manik S (2024) A comprehensive review of the dairy pasteurization process using machine learning models. Food Control 164:110574. https://doi.org/10.1016/j.foodcont.2024.110574

Singhal A, Kumari N, Ghosh P, Singh Y, Garg S, Shah MP, Jha PK, Chauhan DK (2022) Optimizing cellulase production from *Aspergillus flavus* using response surface methodology and machine learning models. Environ Technol Innov 27:102805. https://doi.org/10.1016/j.eti.2022.102805

Sinha R, Singh S, Srivastava P (2014) Studies on process optimization methods for Rapamycin production using *Streptomyces hygroscopicus* ATCC 29253. Bioprocess Biosyst Eng 37:829–840. https://doi.org/10.1007/s00449-013-1051-y

Sodhi AS, Sharma N, Bhatia S, Verma A, Soni S, Batra N (2022) Insights on sustainable approaches for production and applications of value added products. Chemosphere 286:131623. https://doi.org/10.1016/j.chemosphere.2021.131623

Sounderarajan S, Kanakalakshmi K, Reddy PDS, Sethi G, Ayothiraman S (2024) Machine learning approach for the effectual production of a novel esterase and its application in bioremediation of dairy effluent. Process Biochem 144:20–37. https://doi.org/10.1016/j.procbio.2024.05.005

Srivastava RK, Shetti NP, Reddy KR, Aminabhavi TM (2020) Biofuels, biodiesel and biohydrogen production using bioprocesses. A review. Environ Chem Lett 18:1049–1072. https://doi.org/10.1007/s10311-020-00999-7

Stokes JM, Yang K, Swanson K, Jin W, Cubillos-Ruiz A, Donghia NM, MacNair CR, French S, Carfrae LA, Bloom-Ackermann Z, Tran VM (2020) A deep learning approach to antibiotic discovery. Cell 180:688–702. https://doi.org/10.1016/j.cell.2020.01.021

Sultana N, Hossain SZ, Abusaad M, Alanbar N, Senan Y, Razzak SA (2022) Prediction of biodiesel production from microalgal oil using bayesian optimization algorithm-based machine learning approaches. Fuel 309:122184. https://doi.org/10.1016/j.fuel.2021.122184

Sun W, Shahrajabian MH, Lin M (2022) Research progress of fermented functional foods and protein factory-microbial fermentation technology. Fermentation 8:688. https://doi.org/10.3390/fermentation8120688

Sydney EB, Duarte ER, Burgos WJM, de Carvalho JC, Larroche C, Soccol CR (2020) Development of short chain fatty acid-based

artificial neuron network tools applied to biohydrogen production. Int J Hydrogen Energy 45:5175–5181. https://doi.org/10.1016/j.ijhydene.2019.10.128

Taiwo AE, Musonge P, Biofuels (2023) Bioprod Biorefin 17:961–975. https://doi.org/10.1002/bbb.2490

Tokuyama K, Shimodaira Y, Kodama Y, Matsui R, Kusunose Y, Fukushima S, Nakai H, Tsuji Y, Toya Y, Matsuda F, Shimizu H (2021) Soft-sensor development for monitoring the lysine fermentation process. J Biosci Bioeng 132:183–189. https://doi.org/10.1016/j.jbiosc.2021.04.002

Ündey C, Ertunç S, Mistretta T, Looze B (2010) Applied advanced process analytics in biopharmaceutical manufacturing: challenges and prospects in real-time monitoring and control. J Process Control 20:1009–1018. https://doi.org/10.1016/j.procont.2010.05.008

Usmani Z, Sharma M, Awasthi AK, Sivakumar N, Lukk T, Pecoraro L, Thakur VK, Roberts D, Newbold J, Gupta VK (2021) Bioprocessing of waste biomass for sustainable product development and minimizing environmental impact. Bioresour Technol 322:124548. https://doi.org/10.1016/j.biortech.2020.124548

Volk MJ, Lourentzou I, Mishra S, Vo LT, Zhai C, Zhao H (2020) Biosystems design by machine learning. ACS Synth Biol 9:1514–1533. https://doi.org/10.1021/acssynbio.0c00129

Vora LK, Gholap AD, Jetha K, Thakur RRS, Solanki HK, Chavda VP (2023) Artificial intelligence in pharmaceutical technology and drug delivery design. Pharmaceutics 15:1916. https://doi.org/10.3390/pharmaceutics15071916

Walsh I, Myint M, Nguyen-Khuong T, Ho YS, Ng SK, Lakshmanan M (2022) Harnessing the potential of machine learning for advancing quality by design in biomanufacturing. MAbs 14:2013593. https://doi.org/10.1080/19420862.2021.2013593

Walther C, Voigtmann M, Bruna E, Abusnina A, Tscheließnig AL, Allmer M, Schuchnigg H, Brocard C, Föttinger-Vacha A, Klima G (2022) Smart process development: application of machine-learning and integrated process modeling for inclusion body purification processes. Biotechnol Prog 38:e3249. https://doi.org/10.1002/btpr.3249

Walther C, Martinetz MC, Friedrich A, Tscheließnig AL, Voigtmann M, Jung A, Brocard C, Bluhmki E, Smiatek J (2023) Solubilization of inclusion bodies: insights from explainable machine learning approaches. Front Chem Eng 5:1227620. https://doi.org/10.3389/fceng.2023.1227620

Wang J, Wan W (2009) Optimization of fermentative hydrogen production process using genetic algorithm based on neural network and response surface methodology. Int J Hydrogen Energy 34:255–261. https://doi.org/10.1016/j.ijhydene.2008.10.010

Wang G, Haringa C, Noorman H, Chu J, Zhuang Y (2020) Developing a computational framework to advance bioprocess scale-up. Trends Biotechnol 38:846–856. https://doi.org/10.1016/j.tibtech.2020.01.009

Wang Y, Tang M, Ling J, Wang Y, Liu Y, Jin H, He J, Sun Y (2021a) Modeling biohydrogen production using different data driven approaches. Int J Hydrogen Energy 46:29822–29833. https://doi.org/10.1016/j.ijhydene.2021.06.122

Wang Y, Yang G, Sage V, Xu J, Sun G, He J, Sun Y (2021b) Optimization of dark fermentation for biohydrogen production using a hybrid artificial neural network (ANN) and response surface methodology (RSM) approach. Environ Prog Sustain Energy 40:13485. https://doi.org/10.1002/ep.13485

Wang J, Ding S, Da C, Chen C, Wu Z, Li C, Zhou G, Tang C (2023a) Morphology-Based prediction of proliferation and differentiation potencies of Porcine muscle stem cells for cultured meat production. J Agric Food Chem 71:18613–18621. https://doi.org/10.1021/acs.jafc.3c06919

Wang Z, Peng X, Xia A, Shah AA, Yan H, Huang Y, Zhu X, Zhu X, Liao Q (2023b) Comparison of machine learning methods for predicting the methane production from anaerobic digestion of lignocellulosic biomass. Energy 263:125883. https://doi.org/10.1016/j.energy.2022.125883

Wang N, Yang W, Wang B, Bai X, Wang X, Xu Q (2024a) Predicting maturity and identifying key factors in organic waste composting using machine learning models. Bioresour Technol 400:130663. https://doi.org/10.1016/j.biortech.2024.130663

Wang X, Wang Y, Tseng YY, Gao Y, Li K, Wang MH, Wu J, Lay CH, Huo M (2024b) Integration of the grey relational analysis with machine learning for sucrose anaerobic hydrogen production prediction. Int J Hydrogen Energy 68:388–397. https://doi.org/10.1016/j.ijhydene.2024.04.242

Wasalathanthri DP, Rehmann MS, Song Y, Gu Y, Mi L, Shao C, Chemmalil L, Lee J, Ghose S, Borys MC, Ding J (2020) Technology outlook for real-time quality attribute and process parameter monitoring in biopharmaceutical development—A review. Biotechnol Bioeng 117:3182–3198. https://doi.org/10.1002/bit.27461

Whiteman JK, Gueguim Kana EB (2014) Comparative assessment of the artificial neural network and response surface modelling efficiencies for biohydrogen production on sugar cane molasses. BioEnergy Res 7:295–305. https://doi.org/10.1007/s12155-013-9375-7

Wittmann BJ, Johnston KE, Wu Z, Arnold FH (2021) Advances in machine learning for directed evolution. Curr Opin Struct Biol 69:11–18. https://doi.org/10.1016/j.sbi.2021.01.008

Wolfgang P, Arthur M, Sayuri H (2022) Predicting the metabolic condition of a cell culture. https://patents.google.com/patent/JP7092879B2/en?oq=JP7092879B2 Accessed Jan 05

Woodley JM (2020) Towards the sustainable production of bulk-chemicals using biotechnology. New Biotechnol 59:59–64. https://doi.org/10.1016/j.nbt.2020.07.002

Xiaobo Z, Yaodi Z, Jiyong S, Tingting S, Xuetao H, Wen Z, Xiaowei H (2018) Real-time monitoring system and real-time monitoring method for solid fermentation process condition. https://patents.google.com/patent/CN105259827A/en?oq=CN105259827A Accessed 2 Jan 2025

Xie D (2022) Continuous biomanufacturing with microbes—upstream progresses and challenges. Curr Opin Biotechnol 78:102793. https://doi.org/10.1016/j.copbio.2022.102793

Xie Y, Meng WY, Li RZ, Wang YW, Qian X, Chan C, Yu ZF, Fan XX, Pan HD, Xie C, Wu QB (2021) Early lung cancer diagnostic biomarker discovery by machine learning methods. Transl Oncol 14:100907. https://doi.org/10.1016/j.tranon.2020.100907

Xu H (2023) Artificial neural network-based biomass transformation key variable screening method https://patents.google.com/patent/CN113240114B/en?oq=CN113240114B Accessed 2 Jan 2025

Xu RZ, Cao JS, Luo JY, Feng Q, Ni BJ, Fang F (2022) Integrating mechanistic and deep learning models for accurately predicting the enrichment of polyhydroxyalkanoates accumulating bacteria in mixed microbial cultures. Bioresour Technol 344:126276. https://doi.org/10.1016/j.biortech.2021.126276

Xu C, Coen-Pirani P, Jiang X (2023) Empirical study of overfitting in deep learning for predicting breast cancer metastasis. Cancers 15(1969). https://doi.org/10.3390/cancers15071969

Yadav S, Singh V, Mahata C, Das D (2021) Optimization for simultaneous enhancement of Biobutanol and biohydrogen production. Int J Hydrogen Energy 46:3726–3741. https://doi.org/10.1016/j.ijhydene.2020.10.267

Yang S, Tao Y, Maimaiti X, Su W, Liu X, Zhou J, Fan L (2024) Investigation on the exopolysaccharide production from blueberry juice fermented with lactic acid bacteria: optimization, fermentation characteristics and Vis-NIR spectral model. Food Chem 452:139589. https://doi.org/10.1016/j.foodchem.2024.139589

Yeboah MK, Asiedu NY, Dogbe S, Addo A (2024) Performance of machine learning Based-Modelling approach in consolidated

bioprocessing with microbial consortium for bioethanol production. Ind Biotechnol 20:77–97. https://doi.org/10.1089/ind.2024.0003

Zeberli A, Badr S, Siegmund C, Mattern M, Sugiyama H (2021) Data-driven anomaly detection and diagnostics for changeover processes in biopharmaceutical drug product manufacturing. Chem Eng Res Des 167:53–62. https://doi.org/10.1016/j.cherd.2020.12.018

Zhang Y, Jia S, Zhang W (2012) Predicting acetic acid content in the final beer using neural networks and support vector machine. J Inst Brew 118:361–367. https://doi.org/10.1002/jib.50

Zhang W, Li J, Liu T, Leng S, Yang L, Peng H, Jiang S, Zhou W, Leng L, Li H (2021) Machine learning prediction and optimization of bio-oil production from hydrothermal liquefaction of algae. Bioresour Technol 342:126011. https://doi.org/10.1016/j.biortech.2021.126011

Zhang A, Xing L, Zou J, Wu JC (2022) Shifting machine learning for healthcare from development to deployment and from models to data. Nat Biomed Eng 6:1330–1345. https://doi.org/10.1038/s41551-022-00898-y

Zhang L, Zhang C, Zhao X, He C, Zhang X (2024) Improving lipid production by *Rhodotorula glutinis* for renewable fuel production based on machine learning. Front Chem Sci Eng 18:1–13. https://doi.org/10.1007/s11705-024-2410-8

Zheng X (2021) The uncertainties of population research: challenges and opportunities. China CDC Wkly 3:591. https://doi.org/10.46234/ccdcw2021.156

Zhu X, Rehman KU, Wang B, Shahzad M (2020) Modern soft-sensing modeling methods for fermentation processes. Sensors 20:1771. https://doi.org/10.3390/s20061771

Zoffmann S, Vercruysse M, Benmansour F, Maunz A, Wolf L, Blum Marti R, Heckel T, Ding H, Truong HH, Prummer M, Schmucki R (2019) Machine learning-powered antibiotics phenotypic drug discovery. Sci Rep 9:5013. https://doi.org/10.1038/s41598-019-39387-9

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.